Does Fisher–Rao Birth–Death Help Adaptive Biasing Force?
From Model Potentials and a Condensed-Phase Dimer to Molecular Torsions

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Abstract

We study whether augmenting the Adaptive Biasing Force (ABF) method with a Fisher–Rao (FR) birth–death correction improves the computation of a reaction-coordinate free energy, and what kind of improvement it can reasonably be expected to provide. We test this on *five* systems spanning model potentials, a condensed-phase dimer, and molecular torsions: a two-dimensional double-well *metastability* model with a quadrature reference; a two-dimensional *entropic-bottleneck* model with a narrow transverse channel, for which both the reference free energy and the conditional law $Y \mid X = x \sim \mathcal{N}(0, 1/(\beta\omega(x)^2))$ are analytic; a many-body condensed-phase *Weeks–Chandler–Andersen (WCA) dimer* in a dense solvent, evaluated against a high-accuracy thermodynamic-integration reference; and two united-atom alkanes whose slow coordinate is a signed dihedral — *butane* (one torsion, an easy control with an exact analytic reference) and *pentane* (two torsions coupled by the 1–5 “pentane effect”, so that the second dihedral is a hidden slow coordinate when only the first is biased). The alkanes use the correct nonlinear generalized mean force (including the geometric/curvature term) and periodic estimators. In each case the FR step reallocates walkers along the reaction coordinate by killing replicas in over-represented regions and cloning whole configurations from under-represented regions, nudging the empirical marginal $\widehat{p}_t$ toward a target q_t.

The central finding is not a single universal success story, but a regime map. In the metastability model, where plain ABF already samples reasonably well, estimated-target FR is a modest accelerator and the oracle diagnostic shows that target direction can matter. In the cold entropic bottleneck, ABF is starved and estimated-target FR gives a large gain (+51.3%, 20/20 seeds), but the uniform target nearly matches it and the oracle target is worse than ABF; here density correction and free-energy accuracy are not the same objective. A dedicated entropy-dominant stress test (transverse dimension m, total barrier held fixed) refines this further: as the barrier is made more entropic the *achievable* (oracle) gain grows specifically with the entropic share, yet the *deployable* estimated target does not capture it — an entropic-specific benefit that is real but target-limited. In the WCA dimer, tuned FR again improves the free energy reliably (+55.0%, 10/10 seeds), while the best accuracy occurs near the edge of an accuracy–diversity tradeoff. The molecular cases sharpen the picture: in *butane*, ABF already resolves the only slow coordinate, so mFR turns itself off (its birth–death rate falls to ≈ 0.000 and its free-energy error matches ABF, **equivalent** within a pre-registered $\pm 10\%$ practical-equivalence margin) — the extra mechanism has no return where it is not needed, rather than failing. In *pentane*, where a second coupled dihedral makes the conditional law $p(\varphi_2|\varphi_1)$ the binding difficulty, we separate genuine improvement (better $F(\varphi_1)$ with faithful conditional) from false improvement (flatter marginal but corrupted conditional), and report the matched-seed outcome the data support rather than assuming the coupled case must be positive. Across the cases, FR is best understood as a marginal-reallocation mechanism whose value depends on how starved the ABF estimator

is *along the biased coordinate*, how quickly cloned orthogonal degrees of freedom re-equilibrate, and how aggressively birth–death erodes ancestor diversity. The oracle target is therefore a diagnostic control, not a deployable method and not a universal upper bound.

1 Introduction

1.1 Free-energy computation along a reaction coordinate

A central task in computational statistical mechanics is to compute the free energy associated with a low-dimensional *reaction coordinate* of a high-dimensional system in thermal equilibrium [11]. For most of the development we take a configuration to be a point $q = (x, y) \in \mathbb{R}^2$[1] drawn from the canonical (Boltzmann–Gibbs) density

$$\pi(x, y) \ \propto \ e^{-\beta V(x,y)}, \tag{1}$$

where V is the potential energy and $\beta = 1/(k_{\mathrm{B}}T)$ is the inverse temperature. Fixing the reaction coordinate to the first coordinate, $\xi(x, y) = x$, the free energy $F(x)$ and the local mean force $F'(x)$ along ξ are the objects of interest. The difficulty is metastability: when V has well-separated basins along x, unbiased Langevin dynamics crosses the barrier rarely, so a direct estimate of F converges intolerably slowly.

1.2 Adaptive Biasing Force

The Adaptive Biasing Force (ABF) method [5, 6, 8] removes this metastability adaptively. It maintains a running estimate $\widehat{F}'_t(x)$ of the mean force and applies the antiderivative $B_t(x) = \int^x \widehat{F}'_t$ as a bias that depends only on x. As $B_t \to F$, the effective dynamics along x becomes diffusive and the reaction coordinate is explored freely, while — crucially — the bias does not change the conditional law of the orthogonal degrees of freedom given x. ABF is mathematically well understood, with long-time convergence guarantees [10], and is a natural baseline against which to measure any proposed enhancement.

1.3 Why add a Fisher–Rao correction?

ABF accelerates exploration through the *force*, but it does not directly control *where the particles are*. Early in a simulation the estimated bias is poor, so the empirical reaction-coordinate marginal $\widehat{p}_t(x)$ can remain concentrated in the initial basin, starving other values of x of the samples ABF needs to estimate $F'(x)$ there. A Fisher–Rao (FR) birth–death correction [12] addresses exactly this: interpreting the evolution of $\widehat{p}_t$ as a gradient flow of relative entropy in the Fisher–Rao geometry [1], one reallocates probability mass by *killing* particles in over-represented regions and *cloning* particles in under-represented regions, driving $\widehat{p}_t$ toward a chosen target marginal $q_t(x)$ without integrating any additional dynamics. Coupling such a correction to ABF is appealing because the two act on complementary objects (force versus density), but it is not automatically beneficial: a birth–death step resamples *whole* configurations (x, y), and if it distorts the conditional law $Y \mid X = x$ it will corrupt precisely the quantity ABF estimates.

[1] We write a configuration as $q = (x, y)$ in the two-dimensional cases; the many-body case of Section 6 uses $q \in \mathbb{R}^{2n}$ with a scalar collective variable $z(q)$ in place of x (Remark 1). The symbol $q_t(\cdot)$, always carrying a time subscript and a scalar argument, denotes instead the Fisher–Rao *target marginal* of Section 3; the two are never used in the same role.

1.4 Central question and contributions

This report asks how FR changes ABF across *five* controlled systems, not whether one slogan can describe all of them:

1. **When does marginal reallocation help ABF, and when can it hurt?** (Q1)
2. **What target and rate choices make FR useful in each regime?** (Q2–Q3)
3. **What limitations and failure modes appear when the system becomes more realistic?** (Q4)
4. **On a genuinely molecular reaction coordinate (a signed dihedral), does marginal FR help, is it superfluous, or can it produce a "false improvement" that flattens the biased marginal while corrupting a hidden conditional law?** (Q5)

The systems are (i) a two-dimensional double-well *metastability* model with a quadrature reference (Section 4); (ii) a two-dimensional *entropic-bottleneck* model with a narrow transverse channel, for which both the reference free energy *and* the conditional law $Y \mid X = x$ are analytic (Section 5); (iii) a many-body *Weeks–Chandler–Andersen (WCA) dimer* in a dense solvent with a high-accuracy thermodynamic-integration reference (Section 6); and two united-atom alkanes whose slow coordinate is a signed dihedral (Section 7) — (iv) *butane*, an easy control with an exact analytic torsional reference (Section 8), and (v) *pentane*, whose two coupled dihedrals make the second a hidden slow coordinate when only the first is biased (Section 9). Together they span an easy baseline where target direction can matter, a cold analytic bottleneck where density correction and free-energy accuracy can come apart, a many-body system where improved coverage must be balanced against lineage diversity, and two molecular torsions that test the mechanism where the collective variable is a nonlinear function of a three-dimensional configuration.

Our contributions are: (i) a consistent ABF–FR notation that covers the analytic two-dimensional models, the many-body WCA collective variable, and the periodic molecular dihedral with its correct nonlinear generalized mean force (Sections 2, 3 and 7.2); (ii) five matched-seed production studies that quantify both the practical estimated-target method and diagnostic uniform/oracle targets, including a pre-registered practical-equivalence test for the easy butane control and configuration-resolved hidden-conditional diagnostics for pentane (Sections 4 to 6, 8 and 9); and (iii) a cross-case synthesis (Section 10) that explains why the conclusions are intentionally different across examples. A recurring methodological point is the careful interpretation of an *oracle* target that uses the reference free energy: it is a diagnostic control, never a deployable method and not a universal upper bound. Its behaviour is itself part of the evidence: best in the easy metastability setting, approximately tied with the practical target in WCA, and worse than ABF in the cold entropic bottleneck.

2 Background: ABF for a reaction-coordinate free energy

2.1 Free energy and the unbiased marginal

With reaction coordinate $\xi(x, y) = x$, the unbiased reaction-coordinate marginal of the canonical density (1) is

$$p_{\mathrm{ref}}(x) \;=\; \frac{\displaystyle\int e^{-\beta V(x,y)}\,\mathrm{d}y}{\displaystyle\iint e^{-\beta V(x,y)}\,\mathrm{d}x\,\mathrm{d}y}, \tag{2}$$

and the free energy along ξ is, up to an additive constant,

$$F_{\mathrm{ref}}(x) \;=\; -\frac{1}{\beta}\,\log p_{\mathrm{ref}}(x) + C. \tag{3}$$

Because F is defined only up to the constant C, every free-energy profile in this report — both estimates and the reference — is *centred* (its mean over the interior evaluation window is subtracted) before any L^2 error is formed; otherwise a meaningless constant offset would dominate the comparison. The reference quantities are computed by a one-dimensional y-quadrature on a fine grid with a log-sum-exp shift for numerical stability, which is exact up to discretisation and provides the ground truth $F_{\mathrm{ref}}, F'_{\mathrm{ref}}, p_{\mathrm{ref}}$.

2.2 The mean force is the object ABF estimates

Differentiating (3) and using $\partial_x p_{\mathrm{ref}}$ gives the *mean force*

$$F'_{\mathrm{ref}}(x) \;=\; \frac{\displaystyle\int \partial_x V(x,y)\, e^{-\beta V(x,y)}\,\mathrm{d}y}{\displaystyle\int e^{-\beta V(x,y)}\,\mathrm{d}y} \;=\; \mathbb{E}_{\pi}[\,\partial_x V(X,Y)\,|\,X=x\,]\,. \tag{4}$$

Equation (4) is the cornerstone of ABF: the gradient of the free energy equals the *conditional expectation of the instantaneous force* $\partial_x V$ given the reaction coordinate. It is a local, sample-based quantity — no integral of V is required — so it can be estimated on the fly from the particles currently at $X \approx x$, and the free energy is then recovered by quadrature, $\widehat{F}(x) = \int^x \widehat{F}'$.

2.3 Biased dynamics

ABF simulates N overdamped Langevin replicas that share one running mean-force estimate $\widehat{F}'_t(x) \approx F'_{\mathrm{ref}}(x)$ with bias potential $B_t(x) = \int^x \widehat{F}'_t$. Each replica (X_t, Y_t) evolves as

$$\begin{aligned} \mathrm{d}X_t &= \big[-\partial_x V(X_t,Y_t) + \widehat{F}'_t(X_t)\big]\,\mathrm{d}t + \sqrt{2/\beta}\;\mathrm{d}W^x_t, \\ \mathrm{d}Y_t &= -\,\partial_y V(X_t,Y_t)\;\mathrm{d}t + \sqrt{2/\beta}\;\mathrm{d}W^y_t, \end{aligned} \tag{5}$$

with independent Brownian motions W^x, W^y. The added force $+\widehat{F}'_t(X_t)$ acts *only* in the x-direction; the y-dynamics is unbiased. When $\widehat{F}'_t \to F'_{\mathrm{ref}}$ the net drift along x becomes $-\partial_x(V - F)$, whose x-marginal equilibrium is flat, so the reaction coordinate diffuses freely across the barrier.

2.4 Why the bias does not spoil the mean-force estimate

A bias that depends only on x leaves the conditional law of Y given $X = x$ invariant. This is what makes ABF self-consistent.

Proposition 1 (Conditional invariance under an x-only bias)**.** *Let ν be the canonical measure $\nu \propto e^{-\beta V(x,y)}$ and let $\nu_B \propto e^{-\beta\,[V(x,y)-B(x)]}$ be the measure biased by any function B of x alone. Then for every x the conditionals coincide, $\nu_B(y \mid X=x) = \nu(y \mid X=x)$, and consequently*

$$\mathbb{E}_{\nu_B}[\partial_x V(X,Y) \mid X=x] \;=\; \mathbb{E}_{\nu}[\partial_x V(X,Y) \mid X=x] \;=\; F'_{\mathrm{ref}}(x). \tag{6}$$

Proof. At fixed $X = x$ the bias factor $e^{\beta B(x)}$ is a constant in y and cancels between numerator and normaliser of the conditional density:

$$\nu_B(y \mid X = x) = \frac{e^{-\beta[V(x,y)-B(x)]}}{\int e^{-\beta[V(x,y')-B(x)]}\,\mathrm{d}y'} = \frac{e^{-\beta V(x,y)}}{\int e^{-\beta V(x,y')}\,\mathrm{d}y'} = \nu(y \mid X = x).$$

Taking the conditional expectation of $\partial_x V$ under this common conditional gives (6), the last equality being (4). □

Proposition 1 is the reason ABF works: even though the biased ensemble (5) has a different x-marginal than the canonical one, the conditional samples $Y \mid X = x$ remain canonical, so the empirical conditional average of $\partial_x V$ is an unbiased target for $F'_{\mathrm{ref}}(x)$. Any modification of ABF must respect this property; in particular, a birth–death correction that resamples whole configurations (x, y) is only admissible if it too leaves $Y \mid X = x$ (approximately) intact — a hypothesis we test directly in each of the three case studies (Sections 4 to 6).

Remark 1 (Generalisation to many-body systems). Two of our three systems are two-dimensional with $\xi = x$, but the WCA dimer of Section 6 is a many-body system: a configuration is the full vector $q \in \mathbb{R}^{2n}$ of all particle coordinates, and the reaction coordinate is a scalar *collective variable* $z(q)$ (a scaled bond length) rather than a single Cartesian component. Proposition 1 holds verbatim under the substitution $x \mapsto z(q)$ and $y \mapsto$ the orthogonal complement of z on the constant-z manifold (here the solvent coordinates and the transverse dimer modes): a bias $B(z)$ that depends on the collective variable alone is constant on each level set of z, so it cancels in the conditional density and leaves the law of all orthogonal degrees of freedom invariant. Consequently the mean force along z — which for a nonlinear $z(q)$ also carries a geometric term (Section 6) — is still estimated without bias. The only requirement the FR step must meet is the same one as in two dimensions: a clone must copy the *entire* configuration $q \in \mathbb{R}^{2n}$, so that the orthogonal coordinates travel with z and the conditional law is preserved.

2.5 The ABF mean-force estimator

In the kernel form used by the reference implementation, $\widehat{F}'$ is a Nadaraya–Watson running average of the instantaneous force,

$$\widehat{F}'_n(x) \;=\; \frac{\sum_{k\le n}\sum_i K_h(x - X^i_k)\,\partial_x V(X^i_k, Y^i_k)}{\sum_{k\le n}\sum_i K_h(x - X^i_k)}, \tag{7}$$

with a Gaussian kernel K_h of bandwidth h, accumulated over all replicas i and past steps k; the free-energy estimate is its pinned trapezoidal antiderivative, $\widehat{F}_n(x) = \int_0^x \widehat{F}'_n$, and we use $B_t \equiv \widehat{F}$ as the bias. The GPU backend used for the production study replaces the $O(n_{\mathrm{grid}}\,N)$ kernel sums by an $O(N + n_{\mathrm{grid}})$ *binned-smoothed* estimator: particles are scattered onto the x-grid and the count and force histograms are Gaussian-smoothed, so that $\widehat{F}'(x) = \mathrm{smooth(force)/smooth(count)}$ is a Riemann-sum approximation of (7) that converges to it as the grid spacing vanishes. The two estimators are validated to agree to well under 1% on static tests (Section 4.1).

3 The Fisher–Rao marginal correction

The Fisher–Rao (FR) correction is a birth–death / particle-reallocation step that acts on the *reaction-coordinate marginal* of the ensemble. It is interleaved with the ABF dynamics (5) and is designed to drive the empirical marginal $\widehat{p}_t(x)$ toward a target marginal $q_t(x)$, without adding any drift to the (x, y) dynamics.

3.1 Empirical and target marginals

At step n, after the Langevin proposal of all N replicas, the empirical reaction-coordinate marginal is estimated by a (reflected, edge-corrected) kernel density / binned-smoothed estimate of bandwidth η,

$$\widehat{p}_n(x) \;=\; \frac{1}{N}\sum_{i=1}^{N} K_\eta\big(x - X_n^i\big), \tag{8}$$

renormalised on the x-grid. The target $q_n(x)$ is one of three choices (Section 3.6); all are nonnegative and normalised, $\int q_n = 1$.

3.2 Fisher–Rao score

For each particle the FR *score* is the pointwise log-ratio of the empirical to the target marginal, with its ensemble mean removed:

$$S_i \;=\; \log\frac{\widehat{p}_n(X_n^i)+\varepsilon}{q_n(X_n^i)+\varepsilon} \;-\; \underbrace{\int \widehat{p}_n(z)\,\log\frac{\widehat{p}_n(z)}{q_n(z)}\,\mathrm{d}z}_{=\,\mathrm{KL}(\widehat{p}_n\,\|\,q_n)}, \tag{9}$$

where $\varepsilon > 0$ is a small regulariser and the integral baseline equals the Kullback–Leibler divergence $\mathrm{KL}(\widehat{p}_n\|q_n)$ (a grid quadrature; it coincides with the Monte-Carlo baseline $\frac{1}{N}\sum_k \log[\widehat{p}_n(X^k)/q_n(X^k)]$ up to discretisation). Subtracting the baseline makes the score mean-zero, so the birth–death step below conserves the expected particle number. The score is clipped to $[-S_{\max}, S_{\max}]$ for stability. Its interpretation is direct:

- $S_i > 0$: X^i lies where $\widehat{p}_n > q_n$, an *over-represented* region (too many particles);
- $S_i < 0$: X^i lies where $\widehat{p}_n < q_n$, an *under-represented* region (too few particles).

3.3 Birth–death resampling

The FR step approximates the Fisher–Rao gradient flow of $\mathrm{KL}(\widehat{p}\|q)$ as a fixed-N particle reallocation. Over a sub-step of length Δt and at effective rate γ_n^{eff} (Section 3.5), each *over-represented* particle ($S_i > 0$) is marked for death with probability

$$\mathbb{P}(i\ \mathrm{dies}) \;=\; 1 - \exp\!\big(-\gamma_n^{\mathrm{eff}}\, S_i\, \Delta t\big), \tag{10}$$

and particles with $S_i < 0$ supply clone sources. The WCA and metastability backends implement this in the direct death-replacement form: killed slots are filled by clones sampled from the under-represented pool, with weights increasing with $|S_i|$. The entropic-bottleneck backend uses a vectorised but closely related fixed-N survivor/clone-pool variant. This distinction matters for exact diagnostics, but not for the mechanism studied here: all variants move mass along the reaction coordinate by copying whole configurations from under-represented regions into over-represented ones.

Cloning copies the *whole* configuration (X^i, Y^i) — the score depends only on x, but the y-coordinate must travel with its x-coordinate, or Proposition 1 would be violated. The step is made deliberately gentle by three safeguards: (i) a burn-in and soft ramp on the rate; (ii) score clipping to $[-S_{\max}, S_{\max}]$; and (iii) a hard cap $f_{\max}$ on the fraction of particles replaced per event (`max_event_fraction`). The numerical values of $S_{\max}$, $f_{\max}$ and the FR stride `fr_every` are case-specific and are listed in each case's design table.

3.4 Notation across cases

The same FR principle is applied to all three systems, but the implementation and two parameter names differ slightly across backends. The birth–death *rate* is written γ in the analytic two-dimensional cases (Sections 4 and 5) and `fr_rate` in WCA (Section 6); they play the same role in (10). The *onset* of FR is parameterised as a burn-in *fraction* b in the analytic cases ((11), $n_{\text{burnin}} = \lfloor b\, n_{\text{steps}} \rfloor$) and as an absolute step count `fr_start_steps` in WCA; the two are related by $b \approx$ `fr_start_steps`$/n_{\text{steps}}$. Where a case uses a collective variable $z(q)$ rather than x, read $x \mapsto z$ throughout this section (Remark 1).

3.5 Rate schedule

The FR rate is zero during an initial burn-in of $n_{\text{burnin}} = \lfloor b\, n_{\text{steps}} \rfloor$ steps (burn-in fraction b), then ramps softly to its maximum γ,

$$\gamma_n^{\text{eff}} \;=\; \gamma\left(1 - e^{-(n-n_{\text{burnin}})/n_{\text{ramp}}}\right), \qquad n > n_{\text{burnin}}, \tag{11}$$

and the birth–death step is applied every `fr_every` steps. The burn-in lets ABF build a usable mean-force estimate (hence a usable target) before any particles are moved; the ramp avoids a sudden shock to the ensemble.

3.6 Target marginals: estimated, uniform, oracle

The three targets differ only in which density q_n the score (9) is taken against. Recall $B_t = \widehat{F}_n$ is the current ABF bias.

Estimated target (the practical method). A target free energy $\widehat{F}_n^{\text{target}}$ is maintained independently of the bias by an exponential moving average of the ABF estimate, $\widehat{F}_{n+1}^{\text{target}} = (1-r)\,\widehat{F}_n^{\text{target}} + r\,\widehat{F}_{n+1}$, and the target marginal is the *biased* marginal that this target free energy would induce under the current bias,

$$q_n^{\text{est}}(x) \;\propto\; \exp\!\Big[-\beta\big(\widehat{F}_n^{\text{target}}(x) - B_t(x)\big)\Big]. \tag{12}$$

This uses only quantities available at run time and is the method whose practical value we assess.

Uniform target (a naive ablation).

$$q_n^{\text{unif}}(x) \;=\; \text{Unif}(x) \quad \text{on the } x\text{-grid}. \tag{13}$$

This tests whether simply *flattening* the reaction-coordinate marginal is enough. It is a target ablation, not the intended method; it ignores the fact that, under a bias $B_t \neq F$, the canonical biased marginal is not uniform.

Oracle target (a diagnostic control only).

$$q_n^{\text{oracle}}(x) \;\propto\; \exp\!\Big[-\beta\big(F_{\text{ref}}(x) - B_t(x)\big)\Big]. \tag{14}$$

Remark 2 (The oracle target is not an algorithm)*.* Equation (14) replaces the estimated target free energy by the *reference* free energy F_{ref} — exactly the unknown quantity the whole procedure is trying to compute. The oracle target therefore *cannot* be evaluated in any real free-energy calculation. We include it solely as an ideal-information diagnostic: it asks what the FR birth–death step would do if the target marginal were anchored to the reference. This is not a universal upper bound. If oracle FR improves on estimated FR in a given regime, the gap can be read as target-shape headroom; if it ties or loses, the diagnostic instead says that the limiting mechanism is not the online target estimate, and may even be a mismatch between a fixed reference-shaped target and the transient ABF bias. The practical comparison is always

$$\text{ABF only} \qquad \text{versus} \qquad \text{ABF} + \text{estimated-target FR.}$$

3.7 The coupled ABF–FR step

One outer iteration, given the state $\{(X_n^i, Y_n^i)\}$ and accumulators, is:

1. update the ABF accumulators from the current particles and recompute $\widehat{F}_n', \widehat{F}_n = B_t$ (Section 2.5);
2. propose the biased Langevin move (5) for all replicas;
3. if FR is active at this step (past burn-in, on the `fr_every` schedule): form $\widehat{p}_n$ (8) and the target q_n — one of (12), (13), (14) — compute the scores (9), and apply the birth–death step (10);
4. record diagnostics at the evaluation cadence.

The ABF estimator sees the post-resampling ensemble at the next step, which is the route by which improved x-coverage feeds back into a better mean-force estimate. The remainder of the report quantifies whether, and how, this feedback is beneficial.

4 Case I: a two-dimensional metastability model

Our first system is a two-dimensional double well: the regime in which ABF is already a strong baseline, so it sets the lower end of the difficulty spectrum.

4.1 Model potential and reference

We use a two-dimensional model potential with reaction coordinate $\xi(x, y) = x$,

$$\begin{aligned} V(x,y) = 3\, e^{-x^2}\left(e^{-(y-1/3)^2} - e^{-(y-5/3)^2}\right) \;-\; 5\, e^{-y^2}\left(e^{-(x-1)^2} + e^{-(x+1)^2}\right) \\ +\; 0.2\, x^4 + 0.2\,(y-1/3)^4. \end{aligned} \tag{15}$$

As shown in Figure 1, V has two metastable basins near $x = \pm 1$ separated by a barrier near $x = 0$ (with an upper sub-basin near $(0, 5/3)$), so unbiased dynamics is metastable in x. We fix $\beta = 4$. The reference $F_{\mathrm{ref}}, F'_{\mathrm{ref}}, p_{\mathrm{ref}}$ are obtained by the y-quadrature of (2)–(4) on a fine grid and are treated as ground truth.

4.2 Methods, metrics, and study design

Four methods are compared, identical except for the FR target: **ABF only** (no birth–death; baseline); **ABF+FR estimated** (12) (the practical method); **ABF+FR uniform** (13) (naive marginal-flattening ablation); and **ABF+FR oracle** (14) (diagnostic control, Remark 2). All four share the same potential, integrator, ABF estimator, seeds and initial conditions, so differences are attributable to the FR step and its target alone (matched-seed comparison).

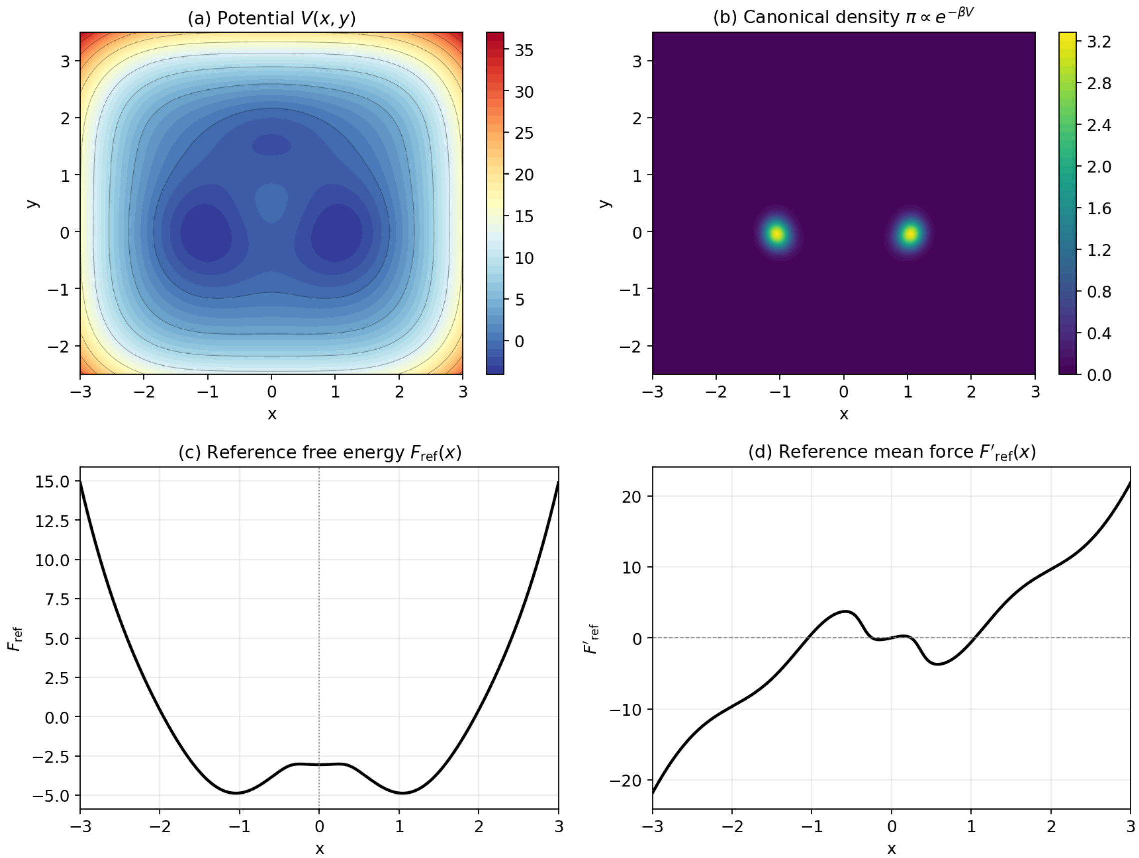


Figure 1: Problem geometry. (a) The potential $V(x, y)$ (15) with two basins near $x = \pm 1$. (b) The canonical density $\pi \propto e^{-\beta V}$ at $\beta = 4$, concentrated in the two basins. (c) The reference free energy $F_{\mathrm{ref}}(x)$ (a double well in x) and (d) the reference mean force $F'_{\mathrm{ref}}(x)$, both by y-quadrature.

4.2.1 Metrics

All profile errors are interior, domain-RMS L^2 distances on the window $x \in [-2.5, 2.5]$ (the reflecting walls and the x^4 confinement make the edges uninformative), with free-energy profiles centred as in Section 2.

Primary (practical) metrics.

$$\left\| \widehat{F}_T - F_{\mathrm{ref}} \right\|_{L^2}, \qquad \left\| \widehat{F}'_T - F'_{\mathrm{ref}} \right\|_{L^2}, \qquad \int_0^T \left\| \widehat{F}_t - F_{\mathrm{ref}} \right\|_{L^2} \mathrm{d}t. \tag{16}$$

The first two are *final-budget accuracy* at the end of the run; the *integrated* error rewards *fast* convergence (anytime performance), since a method that reaches a good profile earlier accumulates less error over $[0, T]$. These two notions need not be optimised by the same hyperparameters, a point we treat carefully in Section 4.3.

Diagnostic metrics. Reaction-coordinate marginal flatness ($\|\widehat{p} - \mathrm{Unif}\|$ and $\|\widehat{p} - q\|$); cumulative barrier crossings at $x = 0$; region fractions (left basin / barrier / right basin); the FR event fraction (particles replaced per FR application) and the FR score dispersion $\mathrm{std}_i\, S_i$; conditional

Table 1: Production study design (read from the result CSVs and `configs/two_dim_xi_x_production_gpu.yaml`).

Quantity	Value
Inverse temperature β	4.0
Time step Δt	0.002
Steps per run	100,000
Particles per run N	1000
Seeds	5 (0, 1, 2, 3, 4)
Methods	4 (ABF; FR estimated / uniform / oracle)
FR rate γ	0.01, 0.02, 0.05, 0.1
FR bandwidth η	0.075, 0.1, 0.15
FR burn-in fraction	0, 0.2, 0.4
FR stride `fr_every`	5
Total configurations	109
Total runs	545

Table 2: Hyperparameters of the selected configurations (integrated-$L^2(F)$ rule). n_{FR} is the Fisher–Rao stride `fr_every`.

Method	target	γ	η	burn-in	n_{FR}	seeds
ABF only	none	0	0.075	0	1	5
FR estimated	estimated	0.1	0.075	0	5	5
FR uniform	uniform	0.05	0.15	0.2	5	5
FR oracle*	oracle	0.1	0.075	0	5	5

$Y \mid X = x$ errors (Section 4.2.1); and ancestor/diversity counts. We stress that *marginal flatness is a diagnostic, not the objective*: the goal is free-energy/mean-force accuracy (16). A method can flatten $\widehat{p}$ and yet fail to improve $\widehat{F}$ if it does so by corrupting the conditional samples.

Conditional $Y \mid X$ diagnostics. At a set of x-bin centres we compare the empirical conditional $\widehat{p}(y \mid x \in I_j)$ (the y-histogram of particles whose x falls in a narrow window) to the reference conditional $p_{\mathrm{ref}}(y \mid x) \propto e^{-\beta V(x,y)}$, via an L^2 distance and a KL divergence. By Proposition 1 a benign FR step should leave these close to the ABF-only values.

4.2.2 GPU implementation and study size

The production runs use a batched PyTorch backend that simulates many runs at once and reuses the reference metrics/diagnostics code verbatim. For speed it uses the binned-smoothed ABF/FR estimator of Section 2.5 rather than the exact kernel sums (7), and a fully vectorised fixed-N birth–death; the two backends agree to within the stated tolerance ($< 1\%$ in practice) on the validation suite. We flag in Section 11 that this binned estimator is a deliberate, but not mathematically identical, surrogate for the kernel notebook implementation. The FR methods sweep the rate γ, the marginal bandwidth η, and the burn-in fraction b at a fixed FR stride; the resulting study (Table 1) comprises 545 runs over 109 configurations and 5 seeds, at 1000 particles and 100,000 steps each, with *no* NaNs in the primary metrics. The selected configurations are listed in Table 2.

Table 3: Main comparison, configurations selected by smallest median *integrated* $L^2(F)$ (convergence speed). Lower is better except P(beats ABF).

Method	$\int_0^T \|\hat{F}_t - F_{\mathrm{ref}}\|\, dt$	$\|\hat{F}_T - F_{\mathrm{ref}}\|$	$\|\hat{F}'_T - F'_{\mathrm{ref}}\|$	P(beats ABF)	FR event frac.
ABF only	15.91	0.0137	0.0718	–	0
FR estimated	14.08	0.0126	0.0771	0.60	1.6×10^{-5}
FR uniform	14.46	0.0125	0.0769	0.80	5.3×10^{-6}
FR oracle*	13.27	0.0113	0.0602	1.00	1.7×10^{-5}

* The oracle target uses the reference free energy F_{ref} (the unknown being computed); it is a diagnostic positive control, *not* a usable algorithm. Per-config medians over 5 seeds; one configuration per method, selected by the smallest median integrated $L^2(F)$.

4.3 Results

Table 3 reports the four methods, each at its best configuration under the *integrated*-$L^2(F)$ rule (convergence speed; see Section 4.3.2). Reading the practical comparison (ABF only versus ABF+FR estimated): the estimated-target FR reduces the median integrated free-energy error from 15.91 to 14.08, a 11.5% reduction, while the *final* free-energy error improves only slightly (from 0.0137 to 0.0126). The birth–death is extremely gentle: the median per-event replacement fraction is 1.6×10^{-5}, of order one particle in 10^5 per FR application. Figure 2 shows the same story over time: the FR curves detach below the ABF-only curve in the early and middle phase of the $L^2(F)$ panel.

4.3.1 Final profiles and target-type ablation

Figure 3 overlays the final mean-force, free-energy and reaction-coordinate-marginal profiles. All practical methods recover F'_{ref} and F_{ref} well; the visible differences between ABF only and ABF+FR estimated at the final budget are small, consistent with the modest final-error improvements. The x-marginal (panel c) is somewhat flatter under FR, as intended, but marginal flatness is a diagnostic, not the goal.

Figure 4 shows the distribution, over the 36 configurations of each target type, of the seed-median errors. For the *integrated* error all three FR target types have medians below the ABF-only baseline, but the spread is wide and some configurations are *worse* than ABF, so FR does not help unconditionally. For the *final* $L^2(F)$ the estimated and uniform medians sit essentially on the ABF baseline, and on the final *mean-force* error they lie slightly *above* it — only the oracle control is consistently below. Flattening the marginal with an imperfect target buys early speed but not final mean-force accuracy.

4.3.2 Two selection rules give different "best" hyperparameters

Because integrated and final errors measure different things, they need not be optimised by the same configuration. Selecting by *integrated* $L^2(F)$ (Table 3) favours fast convergence and selects an aggressive configuration for the estimated target ($\gamma = 0.1$, burn-in 0); selecting by *final* $L^2(F)$ (Table 4) favours final-budget accuracy and selects a gentle, delayed configuration ($\gamma = 0.01$, burn-in 0.4), which improves the final free-energy error by 12.5% and beats ABF on 80.0% of matched seeds. This dependence on the selection rule is itself a finding.

The *uniform* target is competitive here (integrated reduction 9.1%): under a near-converged bias $B_t \approx F$ the canonical biased marginal is already close to uniform, so "flatten $\widehat{p}$" and "match the biased marginal" nearly coincide. The *oracle* diagnostic (Remark 2) achieves the lowest errors

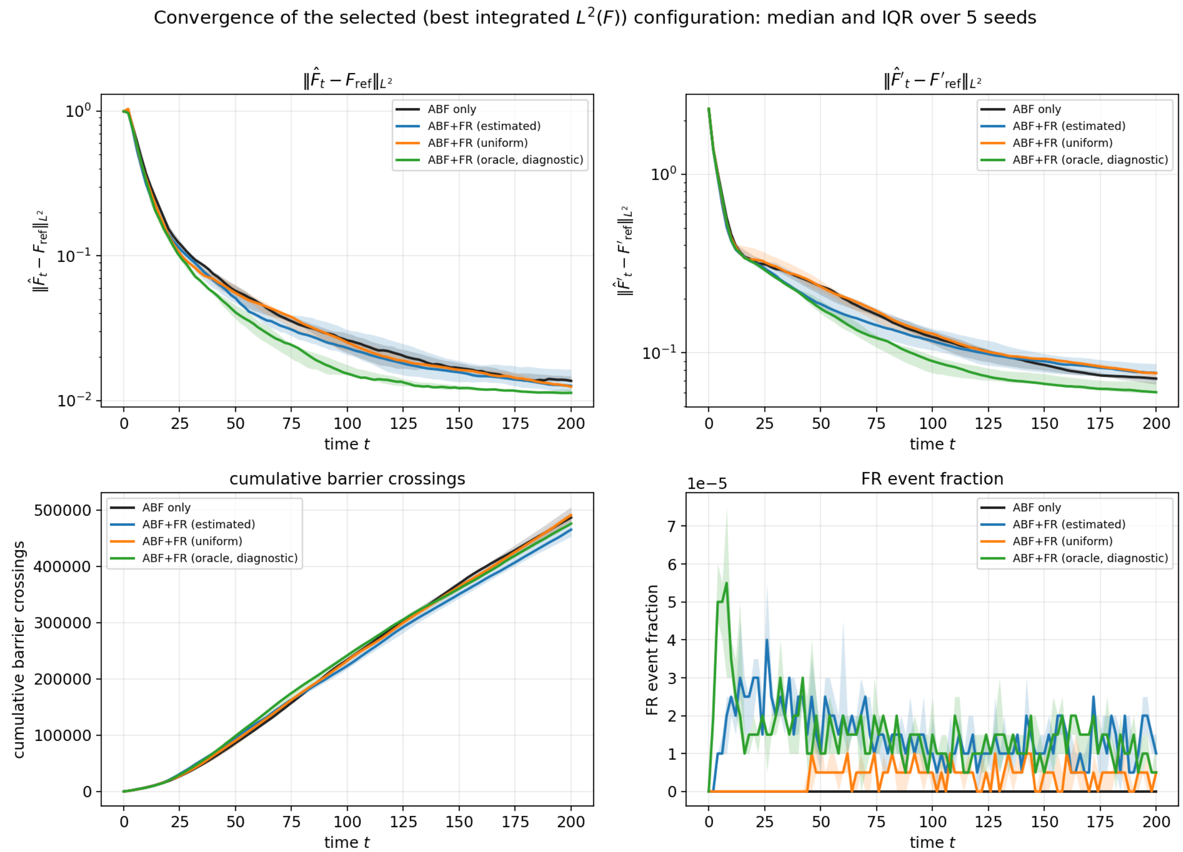


Figure 2: Convergence of the selected (integrated-$L^2(F)$ rule) configuration for each method: median (line) and inter-quartile range (band) over 5 seeds. The free-energy and mean-force errors (top, log scale) fall fastest for the oracle diagnostic and faster for the estimated/uniform FR than for ABF only in the early-to-middle phase; barrier crossings (bottom left) are essentially indistinguishable, and the FR event fraction (bottom right) is of order 10^{-5} throughout. The oracle curve is a diagnostic control (Remark 2).

of all (integrated reduction 16.6%), but it requires F_{ref} and is unusable; the correct reading is comparative: the estimated target captures 11.5% of the reduction against the oracle's 16.6%, so a substantial part of the mechanism's potential is left unrealised by the online target estimate. In this easy regime, then, the oracle–estimated gap localises the practical bottleneck in the *target estimation* — a reading we revisit, and qualify, in Section 10.

4.4 Mechanism

The mechanism is a feedback loop between coverage and the mean-force estimate. ABF needs samples *across* the reaction coordinate: by (4), $F'_{\mathrm{ref}}(x)$ is a conditional average that can only be estimated where particles visit x. Early in a run the bias is poor and the ensemble is unbalanced — Figure 5(b) shows the initial occupancy split ($\sim$0.5 in one basin) relaxing only gradually toward the balanced 1/3–1/3–1/3 split. FR compares $\widehat{p}_t$ with the target q_t and preferentially kills over-represented and clones under-represented particles (10), raising the useful coverage of x earlier than diffusion alone would. Provided the birth–death does not corrupt $Y \mid X = x$ (Proposition 1), better coverage feeds back into a better conditional mean-force estimate.

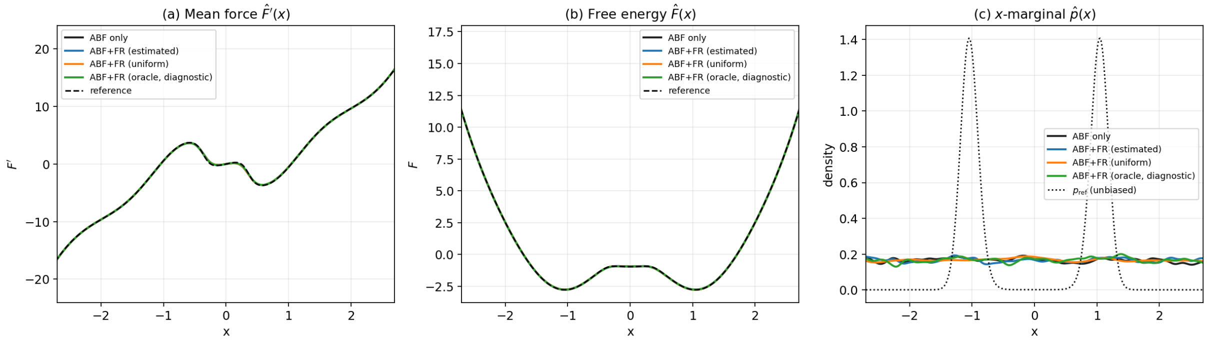


Figure 3: Final profiles (mean over seeds) for the selected configurations. (a) Mean force $\widehat{F}'(x)$ and (b) centred free energy $\widehat{F}(x)$ versus the reference (dashed); (c) reaction-coordinate marginal $\widehat{p}(x)$, with the unbiased p_{ref} (dotted). Practical methods recover the reference well; FR mildly flattens $\widehat{p}$.

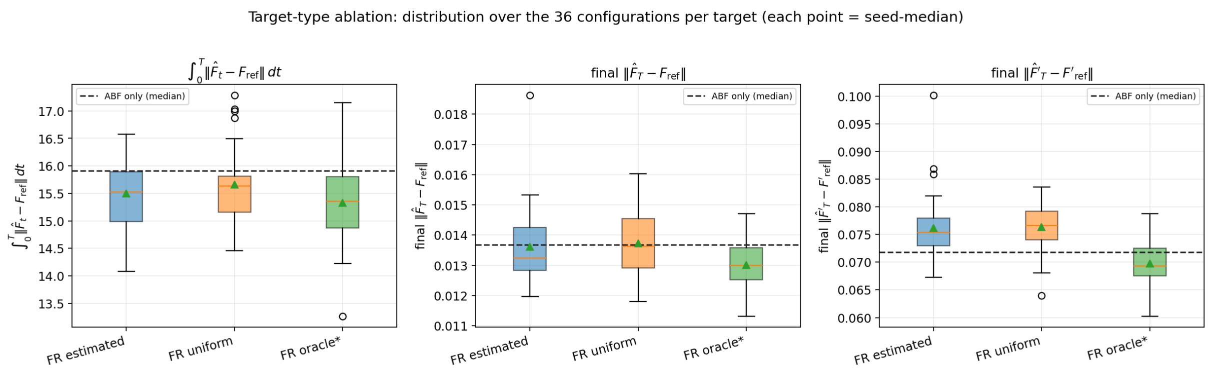


Figure 4: Target-type ablation: each box is the distribution over the 36 configurations of a target type (each point a seed-median); the dashed line is the ABF-only median. FR lowers the *integrated* error (left) but the practical targets barely change the *final* $L^2(F)$ (middle) and slightly raise the final mean-force error (right). Only the oracle diagnostic improves all three.

A notable feature of Figure 5(c) is that FR does *not* produce many more barrier crossings than ABF; the benefit comes from *reallocating* existing replicas toward starved values of x rather than from manufacturing rare transitions. Consistent with this, the event fractions are of order 10^{-5}: in the successful regimes FR is a *weak* correction.

4.4.1 FR does not visibly corrupt $Y \mid X = x$

Figure 6 compares the conditional fidelity of all methods at the final time. The FR variants are *not* worse than ABF only; near the barrier ($x \in \{-0.5, 0, 0.5\}$), where ABF-only under-samples, they are in fact slightly *better* on both the L^2 and KL conditional metrics — direct evidence that the FR mechanism improves x-coverage without corrupting the orthogonal sampling, the benign behaviour Proposition 1 requires.

Remark 3 (Conditional diagnostics are not configuration-resolved). The metastability conditional-diagnostics output records only (method, target type, seed) per row, not the FR hyperparameters. It supports the *aggregate* statement that successful FR does not visibly corrupt $Y \mid X = x$, but cannot isolate a specific "good" versus "bad" configuration's effect. The entropic-bottleneck case (Section 5) closes this gap with an *analytic* conditional law.

Table 4: Supplement: configurations selected by smallest median *final* $L^2(F)$ (final-budget accuracy).

Method	$\int_0^T \|\hat{F}_t - F_{\mathrm{ref}}\|\, dt$	$\|\hat{F}_T - F_{\mathrm{ref}}\|$	$\|\hat{F}'_T - F'_{\mathrm{ref}}\|$	P(beats ABF)	FR event frac.
ABF only	15.91	0.0137	0.0718	–	0
FR estimated	15.25	0.0120	0.0673	0.80	1.1×10^{-6}
FR uniform	14.97	0.0118	0.0680	0.60	7.3×10^{-6}
FR oracle*	13.27	0.0113	0.0602	1.00	1.7×10^{-5}

* Oracle is a diagnostic control (uses F_{ref}), not a deployable method. Configurations here are selected by smallest median *final* $L^2(F)$; cf. Table 3, which need not pick the same hyperparameters.

4.4.2 What determines a good FR?

Figure 7 (estimated target) shows the integrated error decreasing as γ grows — a stronger correction converges faster, which is why the integrated rule selects the aggressive $\gamma = 0.1$ — but the same setting does *not* minimise the final error (Section 4.3.2): there is an explicit speed-versus-accuracy trade-off in γ. The marginal bandwidth η is robust at intermediate values, and an early burn-in helps the integrated error here because the largest imbalance is at early times. Across the successful configurations the event fractions are tiny ($\sim 10^{-5}$) and the cap $f_{\max}$ never binds, so particle diversity is preserved.

4.5 Comparison to OPES

As a correctness gate for the OPES implementation, we run well-tempered OPES [9] on this metastability toy alongside ABF and mFR, using the identical dynamics, mean-force reconstruction, and seeds. On this *easy* case – where plain ABF already converges and the three FR variants agree within seed scatter (Section 4.6) – all methods sit near the same floor: neither mFR nor OPES has headroom to exploit, and the comparison simply confirms that the OPES implementation recovers the correct free-energy shape in the well-sampled basin-and-barrier region $|x| \lesssim 2$ containing both basins and the saddle. No quantitative ranking is claimed here; the decisive comparisons are in Sections 5.7 and 6.13, where ABF is starved and all methods diverge.

4.6 Case finding

On this easy metastability example the answer to Q1 is *yes, but modestly, and chiefly as a convergence accelerator*. Practical estimated-target FR improves the integrated free-energy error by 11.5% with a smaller final-budget gain; the improvement is real but not a rescue, and it is not unconditional. Because plain ABF already converges well here, the room for FR to help is small. The oracle control does best, which — read *within this regime* — points to target estimation as the practical bottleneck; we will see in Sections 5 and 6 that this reading does not survive into the starved regimes, where the oracle's advantage vanishes or reverses.

5 Case II: an entropic bottleneck

The second system tightens a transverse channel at the barrier top and, by lowering the temperature, drives ABF into a genuinely sample-starved regime. It also has a rare luxury: the conditional law $Y \mid X = x$ is *analytic*, so we can test Proposition 1 directly rather than against a histogram.

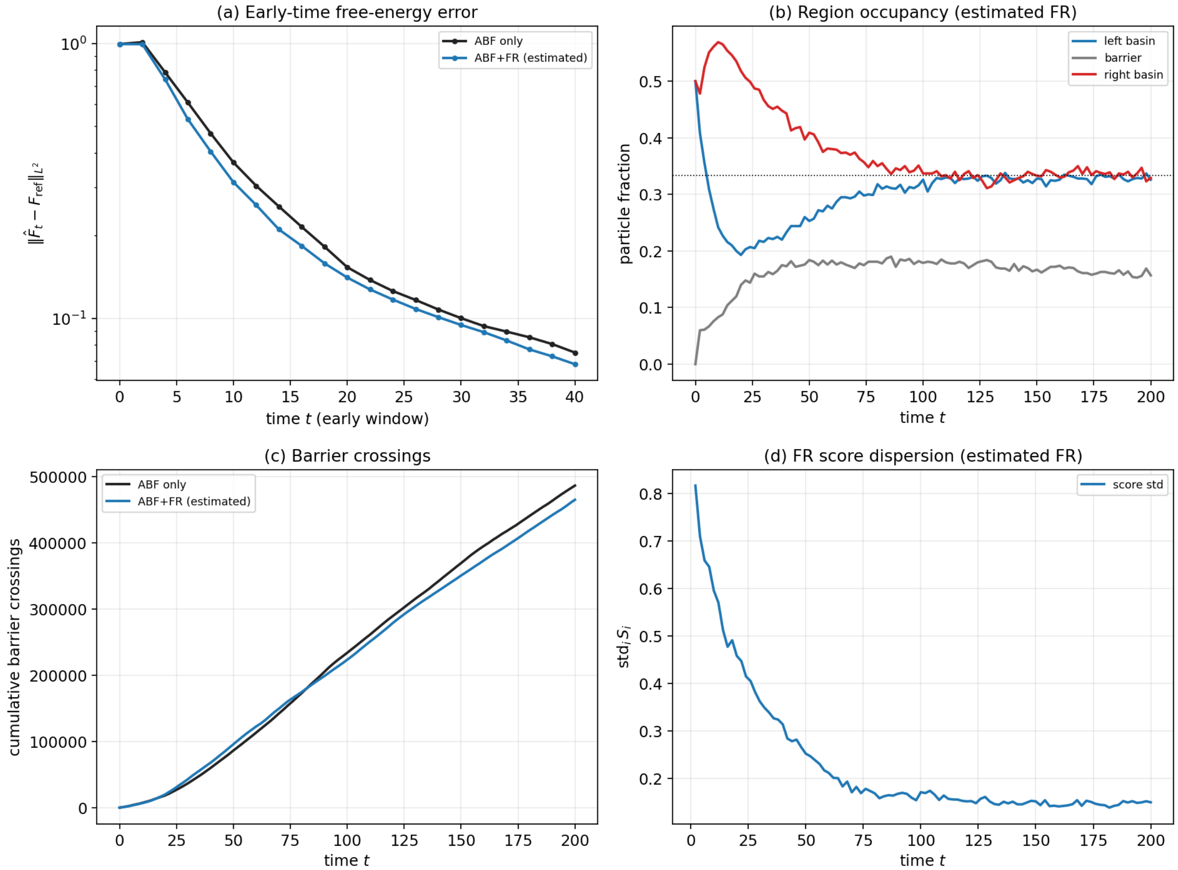


Figure 5: Mechanistic / early-time behaviour, estimated FR (best integrated-rule config) versus ABF only. (a) Early-time free-energy error (log scale): FR converges faster. (b) Region occupancy under FR relaxes from an unbalanced start toward the balanced 1/3 split (dotted). (c) Cumulative barrier crossings are nearly identical — the benefit here is redistribution, not a large increase in raw crossings. (d) FR score dispersion decays as $\widehat{p}_t \to q_t$.

5.1 Model potential and reference

The potential couples a symmetric double well in x to a transverse harmonic channel whose stiffness varies with x,

$$V(x, y) = H\,(x^2 - 1)^2 + \tfrac{1}{2}\,\omega(x)^2\,y^2, \qquad \omega(x) = \omega_{\mathrm{out}} + (\omega_{\mathrm{in}} - \omega_{\mathrm{out}})\,e^{-x^2/2s^2}, \tag{17}$$

with reaction coordinate $\xi(x, y) = x$. The stiffness $\omega(x)$ is large near $x = 0$ (a *narrow channel at the barrier top*) and relaxes to ω_{out} in the wells. Because the y-integral is Gaussian, the reference is analytic up to a constant:

$$F_{\mathrm{ref}}(x) = H\,(x^2 - 1)^2 + \beta^{-1}\log\omega(x) + C, \qquad F'_{\mathrm{ref}}(x) = 4H\,x\,(x^2 - 1) + \beta^{-1}\,\omega'(x)/\omega(x), \tag{18}$$

and, crucially, the conditional law is the exact Gaussian

$$Y \mid X = x \;\sim\; \mathcal{N}\big(0,\; 1/(\beta\,\omega(x)^2)\big). \tag{19}$$

The base regime is $\beta = 8$, $H = 2.5$, $\omega_{\mathrm{out}} = 1$, $\omega_{\mathrm{in}} = 25$, $s = 0.25$, with $N = 256$ walkers, $\mathrm{d}t = 10^{-3}$, and 40 000 steps. The free energy is gauged at $x = 0$ and all L^2 errors are RMS over the interior window $x \in [-1.5, 1.5]$.

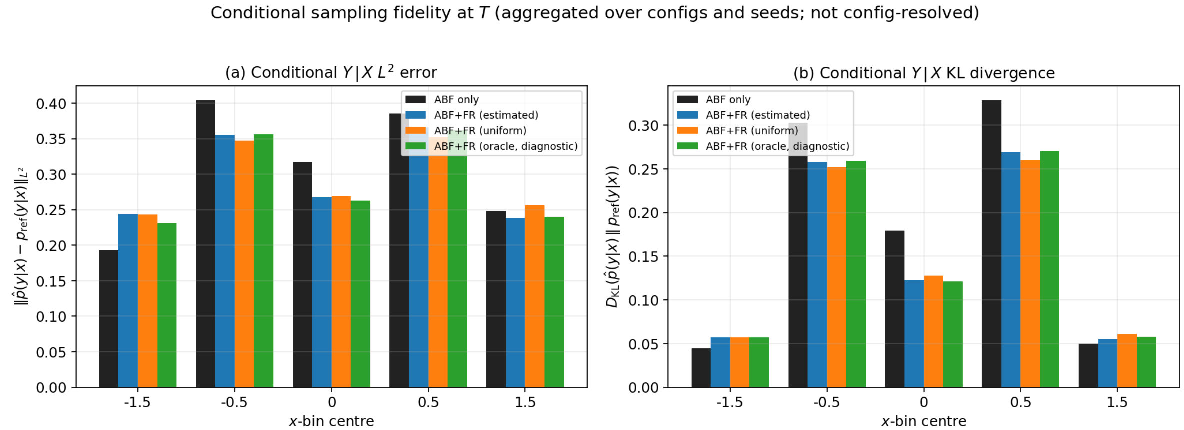


Figure 6: Conditional $Y \mid X = x$ fidelity at T, by x-bin: (a) L^2 distance and (b) KL divergence of $\widehat{p}(y \mid x)$ to $p_{\mathrm{ref}}(y \mid x)$. FR does not inflate the conditional error; near the barrier it slightly reduces it. *Limitation:* the conditional CSV identifies runs only by (method, target, seed) and is aggregated over configurations (Remark 3).

Why this is an entropic bottleneck — and an honest caveat. Near $x = 0$ the accessible y-width shrinks like $1/\omega(x)$: at the base setting the transverse channel is $\omega_{\mathrm{in}}/\omega_{\mathrm{out}} = 25$, i.e. its standard deviation is 25× narrower than in the wells, contributing an *entropic* term $\beta^{-1} \log \omega(x)$ to the free energy. We are explicit, however, that in this parameterisation the entropic contribution $\beta^{-1} \log(\omega_{\mathrm{in}}/\omega_{\mathrm{out}}) \approx 0.40$ is smaller than the energetic barrier height $U(0) - U(\pm 1) = H = 2.5$. The quantity $4H = 10$ is the quartic force prefactor in $F'(x)$, not the free-energy barrier. The bottleneck therefore modulates the transverse width but is *not* the dominant contributor to reaction-coordinate starvation; that role belongs to the energetic double well. This matters for reading the ω_{in} sweep below.

5.2 Methods and design

The practical comparison is again **ABF only** versus **ABF+FR estimated** (12), on matched seeds (identical initial conditions and Langevin noise per seed). The diagnostics **FR uniform** (13) and **FR oracle** (14) (built from the analytic F_{ref}, hence not deployable, Remark 2) are also run. The FR hyperparameters are listed in Table 5; an assertion guards every batched call so the reference never reaches the estimated target. For the ω_{in} sweep only, the stiffest channel is stabilised with a smaller time step and more steps at the same physical time horizon ($\mathrm{d}t = 4 \times 10^{-4}$, 100,000 steps).

5.3 Results

At the base regime the practical method reduces the median final free-energy error from 0.201 (ABF) to 0.098 (ABF+FR estimated), a 51.3% reduction, winning 20 of 20 matched seeds (Figure 8). Two diagnostics are immediately revealing. The *uniform* target — which carries no information about the free-energy shape at all — recovers almost the entire gain (50.5%), while the *oracle* target (the *analytic* F_{ref}) is *worse than plain ABF* (−28.1%). This is the opposite of the metastability ranking (Section 4.6): here a target that over-commits to a fixed shape fights the ABF bias instead of merely equalising coverage. The signature points to *balanced resampling / variance reduction across the reaction coordinate*, not free-energy-shape steering, as the source of the gain.

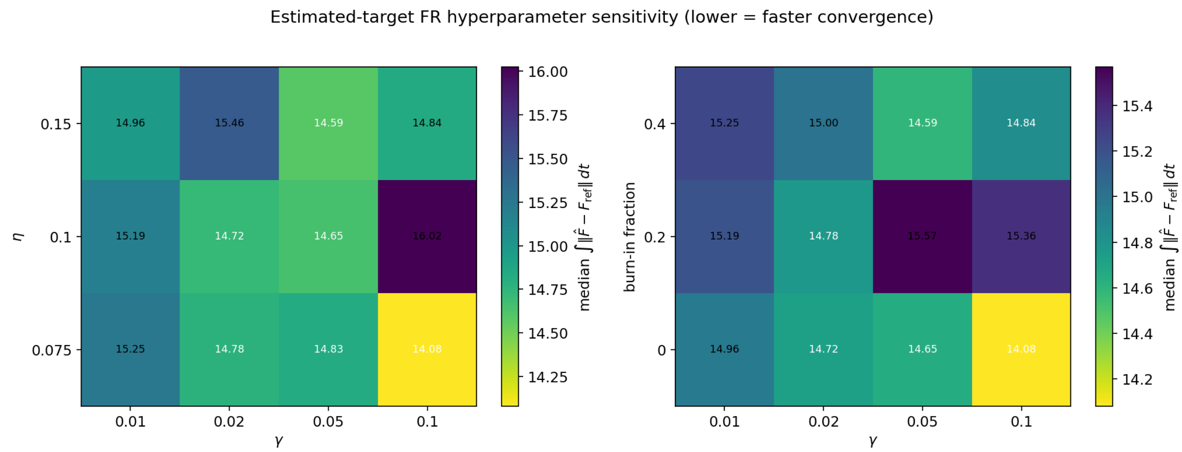


Figure 7: Estimated-target FR hyperparameter sensitivity: median integrated $L^2(F)$ over the grid (lower = faster convergence). Left: γ versus η; right: γ versus burn-in fraction. The integrated error improves with larger γ and earlier burn-in; intermediate η is robust. These trends should not be extrapolated beyond this regime.

Table 5: Entropic-bottleneck base regime and Fisher–Rao hyperparameters.

Quantity	Value
Inverse temperature β	8
Barrier height H	2.5
Channel stiffness $\omega_{\mathrm{out}}, \omega_{\mathrm{in}}$	1, 25
Channel width s	0.25
Walkers N	256
Time step Δt	10^{-3}
Steps per run	40,000
FR rate γ	15
FR stride `fr_every`	10
Score clip $S_{\max}$	3
Event cap $f_{\max}$	0.08
Target EMA rate	0.005

5.3.1 Temperature governs the gain, not the bottleneck width

Varying the inverse temperature β at fixed $\omega_{\mathrm{in}} = 25$ (Table 6) is the cleanest mechanistic signal in the study. FR *helps when sampling is hard and hurts when it is easy*: at high temperature ($\beta = 2, 4$) crossings are frequent, ABF already samples the whole coordinate (its error is ~ 0.05), and FR's birth–death merely injects resampling noise — it is mildly harmful and loses most seeds (-26.1% at $\beta = 4$). At low temperature ($\beta = 8, 12$) crossings are rare, ABF is starved on the far side of the barrier (error 0.2–0.4), and FR's balanced resampling substantially repairs the coverage, winning every seed. The cross-over sits between $\beta = 4$ and $\beta = 8$.

By contrast, sweeping the bottleneck width ω_{in} at fixed $\beta = 8$ (Figure 9) leaves the gain essentially *flat* at $\sim 50\%$. This is the honest consequence of the caveat above: the ABF baseline error is nearly independent of ω_{in} because the ω-independent energetic barrier dominates the sampling difficulty. The gain is large and reliable across ω_{in}, but it is governed by metastability (temperature), not by the bottleneck width. We do not claim a bottleneck-strength scaling the data do not show.

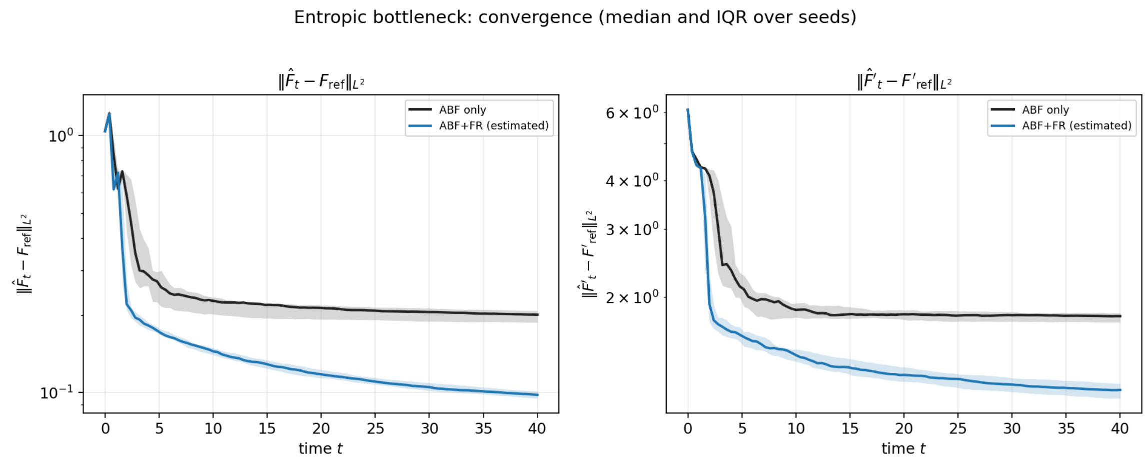


Figure 8: Entropic bottleneck, convergence (median and inter-quartile range over seeds). ABF+FR estimated separates from ABF early and stays below it for the entire trajectory on both the free-energy (left) and mean-force (right) errors.

Table 6: Temperature sweep at $\omega_{\mathrm{in}} = 25$: the FR gain flips sign as ABF crosses from well-sampled (hot) to starved (cold).

β	ABF $\|F - F_{\mathrm{ref}}\|$	FR $\|F - F_{\mathrm{ref}}\|$	gain (%)	win rate
2	0.055	0.060	-8.8	0.30
4	0.049	0.062	-26.1	0.00
8	0.201	0.098	+51.3	1.00
12	0.416	0.281	+32.3	1.00

5.3.2 A wide safe regime for the FR rate

Sweeping the birth–death rate γ at the base regime (Figure 10), the gain increases *monotonically* up to $\gamma = 50$ (+72.6%); we did not reach a failure boundary within the swept range. The windowed ancestor effective sample size falls steadily from 101 to 9 as lineage diversity collapses, yet the free-energy error keeps improving. This differs from the WCA dimer (Section 6), where aggressive birth–death eventually degrades the estimator; the likely reason is the very short conditional-relaxation time of the analytic y-channel here, so a cloned walker re-equilibrates its orthogonal coordinate almost immediately. FR is unusually robust in this model.

5.4 Mechanism: conditional fidelity from an analytic law

Because $Y \mid X = x \sim \mathcal{N}(0, 1/(\beta\omega(x)^2))$ is *analytic* (19), we can test Proposition 1 directly rather than against a noisy histogram — the gap that Remark 3 left open in the metastability case. Figure 11 compares the empirical $\widehat{\mathrm{Var}}(Y \mid X \in I_j)$ to $1/(\beta\omega(x_j)^2)$ at five x-bins. The result is reassuring but not uniformly dominant: FR tracks the analytic variance well across nearly three orders of magnitude (from $\sim 2 \times 10^{-4}$ at the bottleneck $x = 0$ to ~ 0.12 in the wells), and is most competitive in the bottleneck and middle bins, but it is slightly worse than ABF in the well bins for this diagnostic. Thus the free-energy improvement is not coming from a visible collapse of the conditional law; it is achieved while the orthogonal variance remains close to the analytic conditional, with residual errors dominated by finite bin counts rather than a systematic FR bias.

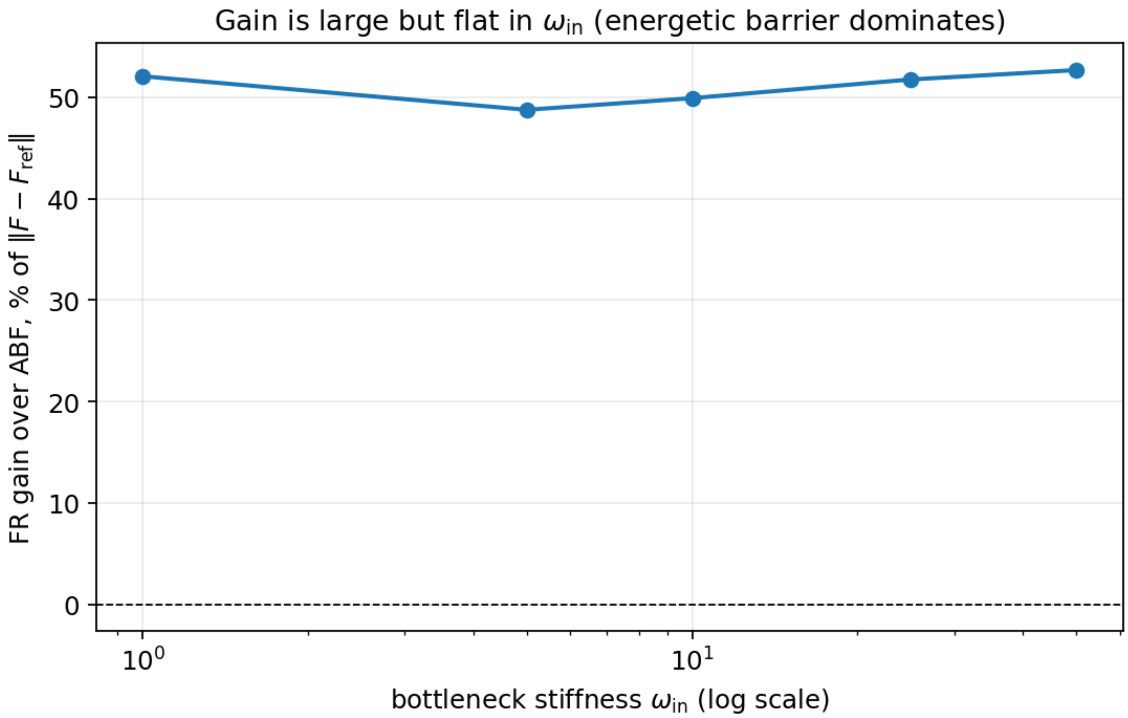


Figure 9: Bottleneck-width sweep ω_{in} at $\beta = 8$. The FR gain is large ($\sim 50\%$) and 100% reliable at every width, but it does *not* grow with the bottleneck strength: the ABF baseline error is flat because the energetic barrier height $H = 2.5$ dominates the smaller entropic contribution $\beta^{-1}\log(\omega_{\text{in}}/\omega_{\text{out}})$.

5.5 An entropy-dominant stress test at fixed total barrier

The caveat of the previous subsections leaves one question open: the gain here tracks *temperature*, but because the energetic barrier dominates we never isolated whether an *entropic* barrier of the same height behaves any differently. We close that gap with a controlled stress test that promotes the transverse coordinate to m dimensions,

$$V(x, \mathbf{y}) = H\,(x^2 - 1)^2 + \tfrac{1}{2}\,\omega(x)^2\,\|\mathbf{y}\|^2, \quad \mathbf{y} \in \mathbb{R}^m, \qquad F_{\text{ref}}(x) = H\,(x^2-1)^2 + \tfrac{m}{\beta}\log\omega(x) + C, \tag{20}$$

so the marginal free energy carries an entropic term $\frac{m}{\beta}\log\omega(x)$ that we can enlarge at will (the conditional law generalises to $\mathbf{Y} \mid X = x \sim \mathcal{N}(0, I_m/(\beta\omega(x)^2))$). Writing the total thermal barrier as $B_0 = \beta H + m\log(\omega_{\text{in}}/\omega_{\text{out}})$ and the *entropic share* $\phi = m\log(\omega_{\text{in}}/\omega_{\text{out}})/B_0$, we set $H = (1 - \phi)B_0/\beta$ and $\omega_{\text{in}} = \omega_{\text{out}}\,e^{\phi B_0/m}$, which holds the total barrier *fixed* while sliding it from purely energetic ($\phi = 0$) to strongly entropic ($\phi = 0.9$). We sweep $\phi \in \{0, 0.25, 0.5, 0.75, 0.9\}$ at $\beta = 4$, $m = 2$, $B_0 = 8$ (20 matched seeds, time horizon $T = 40$); Table 7 lists the design and results.

The outcome is sharper than a simple yes/no, and is the most useful thing this case returns (Figure 12). The *deployable* estimated-target gain does *not* grow with the entropic share: it is U-shaped in ϕ (slope -8.3% per unit ϕ, correlation $r = -0.28$) and is mildly *harmful* in the middle of the sweep (-9.7% at $\phi = 0.5$), where the smooth entropic free-energy bump is hardest to estimate online from the high-variance instantaneous force $\omega(x)\omega'(x)\,\|\mathbf{y}\|^2$. The uniform target behaves almost identically, reaffirming that the practical mechanism is balanced resampling rather than free-energy-shape steering. The *oracle* target tells the opposite story: handed the analytic F_{ref}, its

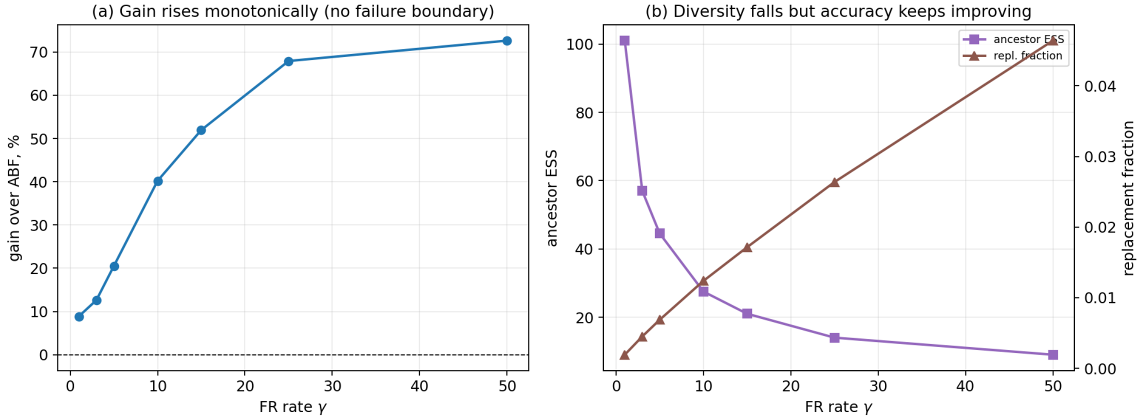


Figure 10: FR-rate sweep $\gamma \in \{1, \ldots, 50\}$ at the base regime. Left: gain over ABF increases monotonically (no failure boundary in range). Right: windowed ancestor ESS collapses ($101 \to 9$) and the replacement fraction rises, yet accuracy keeps improving.

Table 7: Entropy-dominant stress test ($\beta = 4$, $m = 2$, $B_0 = 8$, 20 seeds, $T = 40$): at fixed total barrier, the deployable estimated/uniform gains are small and non-monotone in ϕ, while the oracle gain rises sharply — the entropic-specific headroom is real but target-limited.

ϕ	H	$\omega_{\rm in}$	ABF $\|F - F_{\rm ref}\|$	est. (%)	est. wins	uni. (%)	oracle (%)
0	2.00	1.0	0.0464	+13.0	20/20	+13.0	-6.5
0.25	1.50	2.7	0.0360	+4.7	15/20	+6.0	-5.7
0.5	1.00	7.4	0.0286	-9.7	4/20	-14.3	+9.2
0.75	0.50	20.1	0.0248	-8.1	7/20	-0.5	+43.7
0.9	0.20	36.6	0.0312	+12.7	15/20	+12.9	+42.4

Total barrier $B_0 = \beta H + m \log(\omega_{\rm in}/\omega_{\rm out})$ ($= 8$) is held fixed while ϕ slides it from energetic to entropic ($H = (1-\phi)B_0/\beta$, $\omega_{\rm in} = \omega_{\rm out} e^{\phi B_0/m}$). Gains are matched-seed medians in final $\|F - F_{\rm ref}\|$ vs ABF; the oracle target uses the analytic $F_{\rm ref}$ and is a diagnostic control, *not* deployable.

gain rises steeply with the entropic share (slope 64% per unit ϕ, $r = 0.94$), from harmful at $\phi = 0$ to +43.7% at $\phi = 0.75$. There is therefore a genuinely *entropic-specific* opportunity for birth–death — but it is *target-limited*: realising it needs a free-energy-shaped target accurate enough to steer against, which the online ABF-bias estimate is not in this regime. What the deployable method does deliver across the whole sweep is faster convergence: its integrated-error gain is positive at every ϕ (median +20.5%), and all FR variants preserve the m-dimensional conditional law as in the scalar case.

5.6 Why the deployable gap is an obstruction, not a tuning failure

The stress test leaves a natural hope: if the online ABF-bias estimate is merely *too noisy* to steer against, a better online free-energy estimator inside the target should convert the oracle's entropic headroom into a deployable gain. We tested this hope directly and report a negative result that we believe is the more useful finding, because it is structural. The mean force along the reaction coordinate is

$$\langle f_x \mid x \rangle = 4Hx(x^2 - 1) + \omega(x)\omega'(x)\, \langle \|\mathbf{y}\|^2 \mid x \rangle, \qquad \langle \|\mathbf{y}\|^2 \mid x \rangle = m/(\beta\, \omega(x)^2), \tag{21}$$

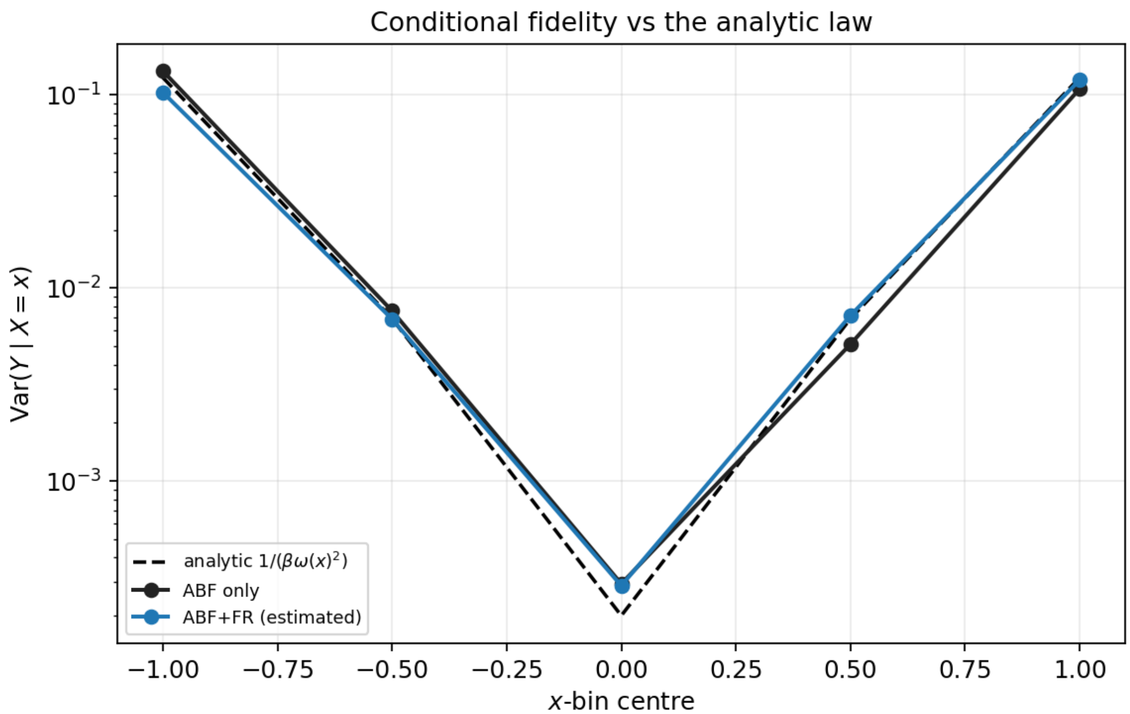


Figure 11: Conditional fidelity against the *analytic* law $\mathrm{Var}(Y \mid X = x) = 1/(\beta\omega(x)^2)$. Empirical FR and ABF variances both track the analytic curve across three orders of magnitude. FR is most competitive near the bottleneck and middle bins, while ABF is slightly closer in the well bins; the diagnostic shows no conditional-law collapse.

and the ABF bias $B(x) = \int^x \langle f_x \mid x' \rangle \, \mathrm{d}x'$ is the sufficient nonparametric estimator of F from the binned per-visit force samples when no external model of the transverse equilibrium is supplied. This is the key reason the entropy-dominant results are not better: the deployable target is not missing a hyperparameter, it is missing information. We constructed several deployable variants that never read F_{ref} and asked whether any beats B as either a readout or an FR target.

(i) Moment fusion. Estimate the entropic part from the transverse log-moment, $F_{\mathrm{ent}}(x) = -\frac{m}{2\beta} \log\langle \|\mathbf{y}\|^2 \mid x \rangle$, and the energetic part from the x-channel. Because the entropic mean force obeys $\mathrm{d}F_{\mathrm{ent}}/\mathrm{d}x = (m/\beta)\, \omega'/\omega = \omega\omega' \langle \|\mathbf{y}\|^2 \mid x \rangle$, the reconstruction telescopes to B *identically* ($\max_x |F_{\mathrm{moment}} - B| = 6 \times 10^{-4}$ on the grid; median moment/ABF readout ratio 0.994 over 20 seeds). The y-channel moment is redundant with the x-force once integrated.

This redundancy also explains why the resulting FR target is weak. The deployable birth–death target has the form

$$q_{\widehat{F}}(x) \propto \exp\{-\beta(\widehat{F}(x) - B(x))\}. \tag{22}$$

If $\widehat{F} = B + \varepsilon$, then $q_{\widehat{F}}(x) \propto \exp\{-\beta\varepsilon(x)\}$ after normalisation. Thus the true free-energy shape cancels out of the practical target: when $\varepsilon \simeq 0$ the target is essentially uniform, and when ε is nonzero the birth–death score steers against estimator noise rather than against the oracle shape. This is the mathematical reason `fr_moment` and `fr_uniform` cluster together.

(ii) Regression intercept. Forcing a non-B signal by regressing f_x on $\|\mathbf{y}\|^2$ per bin (so the slope absorbs $\omega\omega'$) is *worse* than B at every controlled per-bin sample count (ABF/regression RMSE ratio 0.64–0.73, i.e. the regression readout has the larger error), because estimating the slope re-

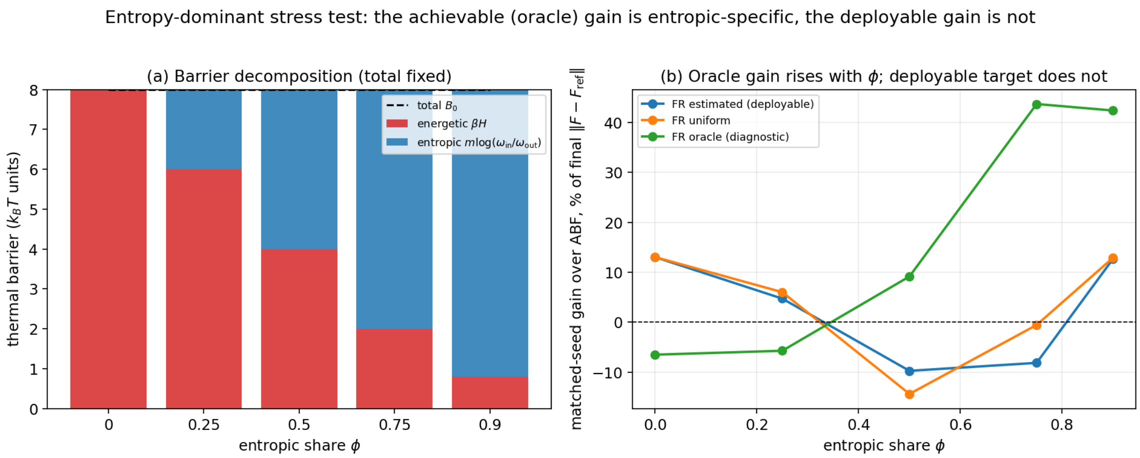


Figure 12: Entropy-dominant stress test at fixed total barrier B_0. (a) The barrier decomposition: as ϕ grows the energetic part βH is traded for the entropic part $m\log(\omega_{\text{in}}/\omega_{\text{out}})$ with the total held constant. (b) Matched-seed gain over ABF: the deployable estimated (and uniform) targets are small and non-monotone in ϕ, whereas the oracle gain rises sharply with the entropic share — the achievable benefit is entropic-specific, but the practical online estimate does not capture it.

injects variance; it also divides by $\mathrm{Var}(\|\mathbf{y}\|^2) \to 0$ in the tight channel and destabilises the feedback loop.

The variance penalty is not accidental. With $u = \|\mathbf{Y}\|^2 \mid X = x$,

$$\mathbb{E}[u \mid x] = \frac{m}{\beta\omega(x)^2}, \qquad \mathrm{Var}(u \mid x) = \frac{2m}{\beta^2\omega(x)^4}. \tag{23}$$

The instantaneous entropic force contribution $\omega(x)\omega'(x)u$ therefore has conditional variance

$$\mathrm{Var}\big(\omega\omega' u \mid x\big) = \frac{2m}{\beta^2}\left(\frac{\omega'(x)}{\omega(x)}\right)^2, \tag{24}$$

and its standard error is reduced only by the number of visits in the bin. Those visits are precisely scarce in the barrier region. Estimators that try to separate the energetic and entropic pieces must estimate additional local slopes or moments from the same scarce samples, so they trade the ABF sample mean for a higher-variance object.

(iii) Smoothing and confidence gating. Wide-smoothing the energetic residual $B - F_{\text{ent}}$ can break the identity $F_{\text{moment}} \equiv B$, but only by introducing a deterministic bias in the target. Because Equation (22) is exponential in this error, a biased $\widehat{F}$ creates a biased cloning/deletion field rather than a harmless post-processing error. Per-bin confidence gating has the opposite pathology: it shrinks the very score that a successful target would need near the bottleneck. In the confirmation sweep the gated target is worse than the ungated moment target, which indicates that the removed component was not reliable shape information but noisy steering.

(iv) Two-dimensional control variate. Stratifying samples by the transverse coordinate $\|\mathbf{y}\|^2$ and recombining per-stratum mean forces — the textbook "resolve the orthogonal coordinate" device — is 2.6× more accurate than plain ABF *when the stratum weights are the analytic χ^2 law*, but reduces to plain ABF *identically* once the weights are estimated from the same data:

$$\sum_k \frac{n_k}{n}\bar{f}_k = \sum_k \frac{n_k}{n}\frac{1}{n_k}\sum_{i\in k} f_i = \frac{1}{n}\sum_i f_i. \tag{25}$$

The measured difference is 3×10^{-16} (machine precision). The entire gain is the injected conditional law, not the stratification.

The same variance logic applies to density- or CZAR-style free-energy targets. If a bin of width h has n_x effective visits, a density estimate has relative noise of order $n_x^{-1/2}$, so the log-density free energy has $\mathrm{Var}(\widehat{F}(x)) = O((\beta^2 n_x)^{-1})$. A steering target, however, uses the *spatial shape* of this estimate. Differencing a noisy log-density to obtain a score or mean-force-like signal adds powers of h^{-1}, while an extended-system restraint contributes its own force variance. Thus the estimator can be asymptotically sensible but still be a bad online control signal at the finite counts available in the bottleneck.

All variants collapse to one statement: any deployable rearrangement of the run's own observables either returns B exactly or departs from it only by added bias or estimation variance, so the steering signal $F_{\text{target}} - B$ is noise rather than shape. The oracle's advantage — and the control variate's analytic-weight advantage — is *information that is not in the trajectory* (the true F_{ref}, equivalently the exact transverse equilibrium). There is also a chicken-and-egg barrier specific to steering: learning that the bottleneck is a high-F region requires *sampling* it, the same starvation that limits B; $\langle \|\mathbf{y}\|^2 \mid x\rangle$ carries no earlier information than the integrated force. A 20-seed confirmation bears this out in dynamics: the moment target (`fr_moment`) and a per-bin confidence-gated variant (`fr_moment_gated`) both sit in the ABF/uniform band (-5.6% and -19.8% at $\phi = 0.75$; 15.2% and 8.4% at $\phi = 0.9$), far from the oracle's $+43.7\%$, and gating *removes* rather than improves the signal — confirming it was noise. The practical consequence is that the entropic-specific headroom is reachable only by *adding information* (enhanced sampling of a genuinely slow orthogonal coordinate — not the case here, where $\mathbf{y}$ equilibrates fast — multiple-replica or tempered data, or a parametric prior on $\omega(x)$ fit from the well interiors), not by a cleverer estimator on the same trajectory.

5.7 Comparison to OPES

As in the WCA case (Section 6.13) we check the starved-regime claim against a modern adaptive-bias method: well-tempered OPES [9] on the identical bottleneck dynamics (β=8, ω_{in}=25), reconstructed by the same mean-force integration and evaluated on the same interior window, with the deposition barrier tuned on held-out seeds and fixed to 4 kT. OPES reaches a median $L^2(F)$=0.3252 over five seeds, against ABF's 0.2098 and mFR's 0.0954: it is *worse than plain ABF* here (-55.0%), and roughly 3.4× less accurate than mFR (-240.9%). The reweighting effective sample fraction is low (n_{eff}=0.0871), consistent with a bias that flattens the x-marginal aggressively but leaves the *conditional* $\mathbf{y}$-average — the entropic part of the mean force — as noisy as under ABF. This is the same lesson as Section 5.6: the entropic obstruction is about *which* average enters the force, and flattening the sampled coordinate does not address it, whereas mFR's ensemble reallocation does. OPES does not close the gap that mFR closes.

5.8 Case finding

The entropic bottleneck is a controlled bridge between the easy metastability case and the hard WCA dimer. Estimated-target FR robustly and substantially reduces the free-energy error in the cold, sample-starved regime ($\sim$ 51.3%, 20/20 seeds), but the gain is governed by *temperature* (FR helps when ABF is starved, mildly hurts when it is not), not by the bottleneck width. The mechanism is balanced resampling / variance reduction, not shape steering: the uniform target nearly matches the estimated one and the oracle target is *worse* than ABF — the reverse of the metastability ranking. The safe FR regime is wide (no failure boundary up to $\gamma = 50$), and the analytic conditional diagnostic shows that the gain is obtained without a systematic collapse of $Y \mid X = x$, even though FR is not uniformly better than ABF on every conditional-variance bin. Finally, the dedicated entropy-dominant stress test (Section 5.5) settles the bottleneck-versus-temperature question that the ω_{in} sweep could not: holding the *total* barrier fixed and growing only the entropic share, the deployable gain still does not increase (it is U-shaped in ϕ), whereas the oracle gain rises sharply to $+43.7\%$ at $\phi = 0.75$. The entropic-specific benefit is therefore real but *target-limited*, and Section 5.6 shows the limit is an *obstruction rather than a tuning failure*: every deployable estimator built from the trajectory's own observables (moment fusion, regression/control variates, smoothing, and confidence gating) reduces to the ABF bias or to the ABF bias plus noise, so the oracle is a provable *information* upper bound that no online estimator on the same data can reach.

6 Case III: a condensed-phase WCA dimer

The third system is many-body and is where ABF is most starved: a dimer that must push a dense solvent aside to change its bond length. It tests whether the two-dimensional findings survive in a setting with thousands of degrees of freedom and a nonlinear collective variable.

6.1 Model and reference

A dimer of two tagged particles sits in a periodic two-dimensional bath of $n_{\text{dim}}^2 - 2$ Weeks–Chandler–Andersen (WCA) solvent particles ($n_{\text{dim}} = 10$, lattice spacing a, box $L = n_{\text{dim}}\, a$). All particles interact through the purely repulsive WCA potential [14] (Lennard–Jones cut and shifted at its minimum $r_0 = 2^{1/6}\sigma$). The two dimer particles additionally feel a symmetric double-well bond $V_{\text{dim}}(r) = h\,(1 - ((r - r_0 - w)/w)^2)^2$ with barrier height $h = 2$ and width $w = 2$. The reaction coordinate is the scaled bond length

$$z(q) = \frac{|q_1 - q_2| - r_0}{2w}, \tag{26}$$

a nonlinear collective variable of the full configuration $q \in \mathbb{R}^{2n}$ (Remark 1): $z \approx 0$ is the *compact* state, $z \approx 1$ the *stretched* state, and $z \in [0.25, 0.75]$ the *transition* region. The free-energy barrier near $z = 0.5$ is partly entropic / crowding-driven, since the dimer must displace solvent to cross it. Dynamics are overdamped Langevin at $\beta = 1$, $\mathrm{d}t = 0.002$, with the minimum-image convention; the default system is $a = 1.5$, $\sigma = \epsilon = 1$. The local mean force along z is the constrained projection of the physical forces, including the geometric contribution with the sign convention of the estimator (Remark 1). The reference $F(z)$ is computed once per system a by constrained-MD thermodynamic integration [3, 7]; it is a high-accuracy numerical reference used for evaluation only, not an analytic exact solution. Errors are taken on the interior window $z \in [-0.1, 1.1]$ with the additive constant aligned.

6.2 Methods and design

As before the practical comparison is **ABF only** versus **ABF+FR estimated** (12), with **FR uniform** and **FR oracle** diagnostics; here the oracle target is built from the TI reference and is explicitly *not* deployable (Remark 2). A no-leakage assertion ensures only the oracle method ever receives the reference, and a clone copies the *entire* WCA configuration — not just z — so cloned solvent arrangements travel with the bond length. Replica lineage is tracked to report the ancestor effective sample size $\text{ESS} = 1/\sum_a w_a^2$. The tuned FR configuration is given in Table 8; the rate is named `fr_rate` and the onset `fr_start_steps` (Section 3.4).

Table 8: WCA dimer base system and tuned Fisher–Rao hyperparameters.

Quantity	Value
Inverse temperature β	1
Time step Δt	0.002
Steps per run	250,000
Replicas N	1024
Lattice spacing a	1.5
Dimer barrier h, width w	2, 2
FR rate `fr_rate`	0.10
FR onset `fr_start_steps`	20,000
FR stride `fr_every`	5
Score clip $S_{\max}$	2.0
Event cap $f_{\max}$	0.02
Target EMA rate	0.005

Table 9: WCA main comparison (250k steps, $N = 1024$, $a = 1.5$). The tuned row is the practical headline; the strong row shows the accuracy–diversity tradeoff. The mean-force gain is modest.

Method	$\|F - F_{\text{ref}}\|$	$\|F' - F'_{\text{ref}}\|$	gain (%)	wins	anc. ESS
ABF only	0.0835	0.3220	–	–	–
FR estimated (tuned)	0.0361	0.2513	+55.0	10/10	86
FR uniform	0.0372	0.2525	+54.6	10/10	90
FR oracle*	0.0387	0.2475	+52.7	10/10	100
FR estimated (strong)	0.0329	0.2590	+58.9	10/10	45
FR estimated (aggressive)	0.0935	0.3883	-13.1	1/10	22

* The oracle target uses the thermodynamic-integration reference F_{ref}; it is a diagnostic control, *not* a deployable method. Medians over 10 seeds.

6.3 Results

The main comparison (Table 9; 10 seeds, $N = 1024$, 250 000 steps) shows the practical method **reliably beating ABF**: the tuned FR cuts the median final $L^2(F)$ from 0.0835 to 0.0361 (+55.0%), winning all 10/10 matched seeds. As in the entropic-bottleneck case, the uniform target (+54.6%) and the oracle target (+52.7%) are *essentially as good as* the estimated target — the online estimated target already extracts nearly all the available benefit, and there is no headroom to the oracle. The stronger estimated-target row in the same table attains the lowest final free-energy error, but with a much smaller ancestor ESS; we therefore use the tuned row as the practical headline and the strong row as the accuracy–diversity contrast. We are explicit that the headline gain is in the *free energy*: the mean-force error $L^2(F')$ improves only modestly ($\approx 22\%$), so we do not claim a large mean-force improvement.

6.4 Mechanism: support reallocation with a diversity cost

The diagnostics support the same broad mechanism as the entropic bottleneck rather than a stationary-coverage reshaping one. Final reaction-coordinate region fractions and the kernel-weighted ABF denominator in the transition window are nearly identical across ABF, FR estimated, FR uniform and FR oracle: FR does *not* converge to a different stationary z-distribution. Instead the gain appears immediately after FR activates and is sustained for the rest of the run. Figure 15 shows the interpretation: birth–death acts as a *support-reallocation resampler for the ABF estima-*

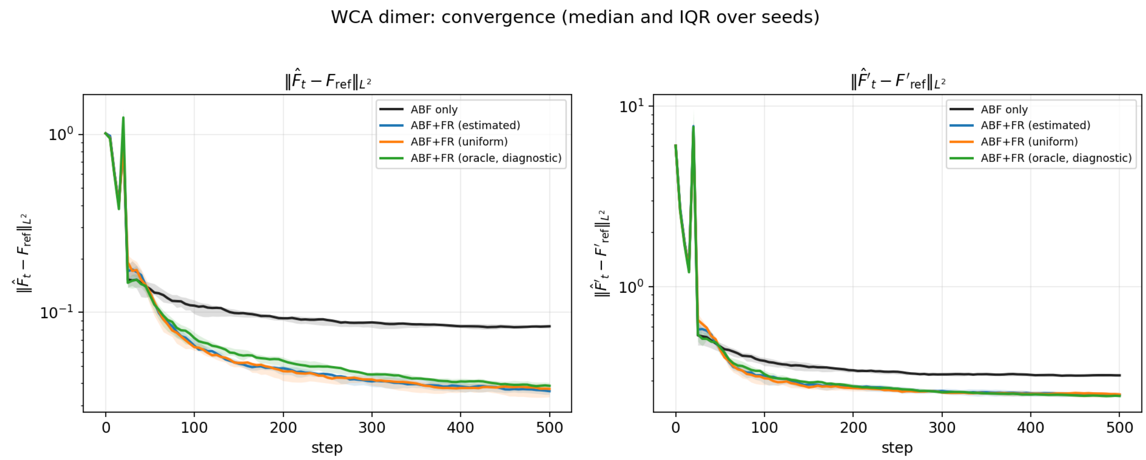


Figure 13: WCA dimer, convergence (median and inter-quartile range over seeds). ABF and FR are identical until FR activates; from then on FR is consistently below ABF on $L^2(F)$ for the rest of the run. The cumulative replacement count (right) confirms the birth–death is active but bounded.

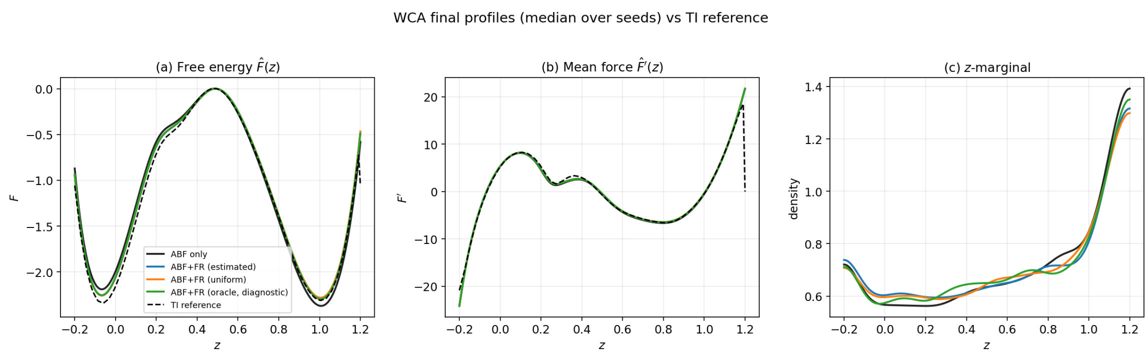


Figure 14: Final profiles (median over seeds) versus the TI reference (dashed): (a) free energy $\widehat{F}(z)$, (b) mean force $\widehat{F}'(z)$, (c) reaction-coordinate marginal. FR recovers the reference free energy more accurately than ABF, chiefly in and around the transition region.

tor. Replicas stuck in over-represented configurations are replaced by whole-configuration clones of under-represented ones, improving weighted force-sample support in important z-bins even when the marginal looks unchanged. A clone is not an independent solvent arrangement at birth; the point is that useful whole configurations are moved into starved reaction-coordinate regions and then decorrelated by subsequent Langevin dynamics. This explains why target shape matters little here, consistent with uniform ≈ estimated ≈ oracle.

6.5 Failure boundary

Unlike the entropic bottleneck, the WCA dimer exhibits a clear failure boundary in the FR rate (Figure 16). The rate ladder (a shorter 150k-step, 4-seed stage) maps a clean safe regime: too gentle (`fr_rate`≤ 0.05) gives a small gain (+19–29%); the sweet spot (`fr_rate`≈ 0.1–0.2) gives the maximal +58–64% with healthy ancestor ESS; the gain then *erodes* sharply (`fr_rate`$= 0.5$ falls to +12% with the error already climbing back toward the ABF baseline) and finally *reverses* for `fr_rate`≥ 1, where the median final $L^2(F)$ is worse than ABF (−38% at rate 1, −68% at rate

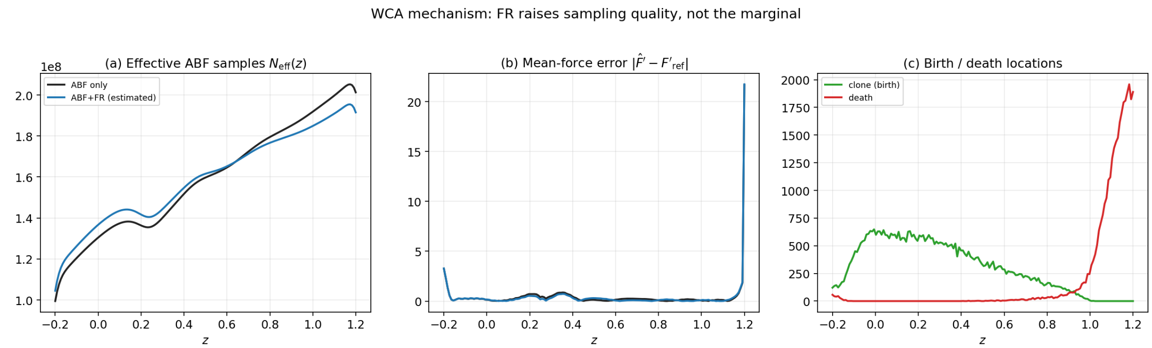


Figure 15: Mechanism. (a) The kernel-weighted ABF denominator $N_{\text{eff}}(z)$ is raised by FR in the transition window; (b) per-bin mean-force error is reduced there; (c) birth (clone) and death locations concentrate around the over/under-sampled regions; FR reshapes the *sampling support*, not the stationary marginal.

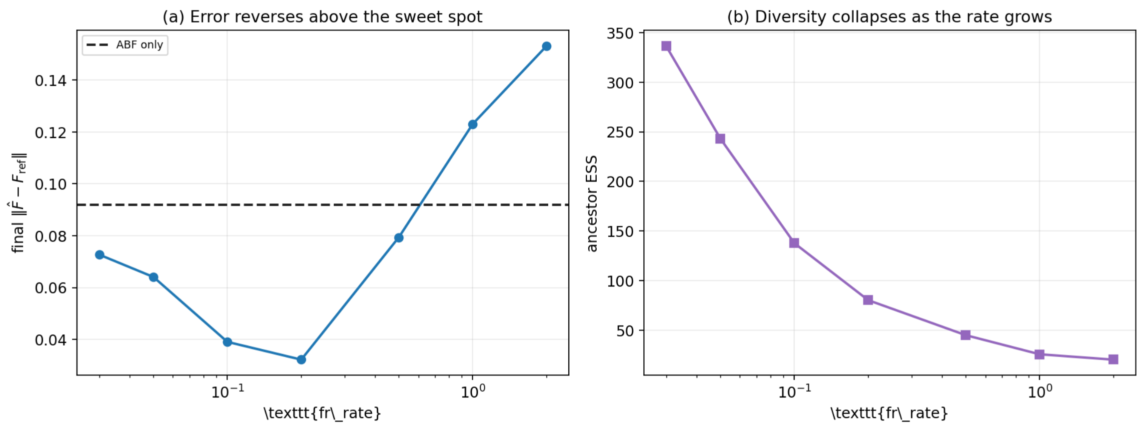


Figure 16: FR-rate ladder (150k steps, 4 seeds). Final $L^2(F)$ (left) is minimised in the 0.1–0.2 sweet spot, climbs back toward the ABF baseline (dashed) by `fr_rate`= 0.5, and rises above it for `fr_rate`≥ 1; ancestor ESS (right) collapses as the rate grows, pinpointing diversity loss as the failure mechanism.

2) and no seed wins, as the ancestor ESS collapses from ~ 337 to ~ 20. The reversal arrives even earlier under a longer budget: the production aggressive configuration (`fr_rate`= 0.5 at 250k steps, 10 seeds) is already net-harmful (−13.1%, winning only 1/10 seeds, ancestor ESS ~ 22), because the extra steps give diversity loss more time to dominate. Over-aggressive birth–death destroys the replica diversity the mean-force estimator depends on, so the estimator degrades faster than coverage improves. This is the predicted failure mode; the analytic y-channel of the entropic bottleneck masks it (Section 5) because cloned walkers re-equilibrate instantly, whereas a dense solvent does not.

6.6 Difficulty dependence

Varying one difficulty axis at a time (Figure 17) shows the gain is robust and, importantly, *grows with difficulty*. With *budget*, FR wins at every n_{steps} and the relative advantage grows rather than washes out (+35.5% at 50k to +56.9% at 400k): ABF's error plateaus around 0.08 after $\sim$ 150k

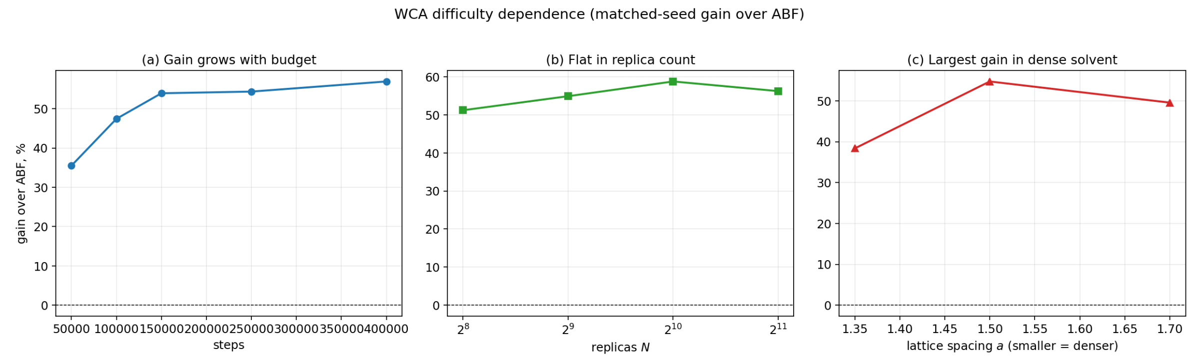


Figure 17: Difficulty dependence (matched-seed gain over ABF). (a) Budget: the relative gain grows with n_{steps} as ABF plateaus and FR keeps improving. (b) Replica count: gain is roughly flat. (c) Solvent crowding: smaller a (denser) yields the largest absolute error reduction. FR wins 5/5 seeds in every cell.

while FR keeps descending, so FR is not merely an early-time accelerator that ABF later catches. With *replica count*, the gain is roughly flat (+51–59% from $N = 256$ to 2048) — FR helps even at $N = 256$. With *solvent crowding*, the densest bath ($a = 1.35$) makes the entropic barrier severe and ABF $\sim 4\times$ worse in absolute error; FR still wins 5/5 and delivers by far the largest absolute error reduction ($0.365 \to 0.225$). In the units a practitioner cares about — absolute free-energy error removed — the crowded regime is where FR helps most.

6.7 WCA phase diagram: when does marginal Fisher–Rao help?

A single WCA setting is not enough to tell whether the gain above is a generic property of the dimer or a response to sample starvation. We therefore added an additive, command-line phase-diagram study that sweeps physical parameters while keeping the existing production result intact. In the production phase diagram reported here we vary the inverse temperature $\beta \in \{1, 2, 4\}$ and dimer barrier height $h \in \{1, 2, 4, 6\}$ at fixed width $w = 2$, lattice spacing $a = 1.5$, and $M = n_{\text{dim}}^2 = 100$ physical particles; we also add a targeted physical-size slice $M \in \{49, 100, 196\}$ at $(\beta, h) = (2, 4)$. This M is the number of physical particles in one WCA system; it is distinct from the $N = 1024$ ABF/FR replicas used to estimate the mean force. Each cell uses 4 matched seeds and 120000 steps. The deployable comparison is ABF versus mFR–ABF with the online estimated target; the uniform target is an ablation, and the oracle target is included as a diagnostic positive control in the table but is not a usable method because it reads the TI reference.

The result is sharper than a monotone "larger βh is harder" story. The measured ABF error, not the nominal product βh, is the useful starvation diagnostic. In the $\beta = 1$ row ABF remains at $L^2(F) \approx 0.09$–0.10, and mFR improves all matched seeds with +47–51% median gain; the anchor cell $\beta = 1, h = 2$ improves from 0.0918 to 0.0474 ($R = 1.94$). At $\beta = 2$ and $\beta = 4$, ABF is often much more accurate (below $L^2(F) = 0.005$ in the easiest accurate cell), and the same birth–death configuration over-corrects: only 5/12 cells have positive gain, and the worst cell β=4, h=1 loses strongly. Thus the phase diagram supports the conditional claim rather than an unconditional one: mFR helps when the ABF free-energy estimator is sample-starved, but it can hurt when ABF has already solved the marginal problem.

The mechanism diagnostics separate exploration from target-shape and genealogy effects. The uniform target is close to the estimated target in this production run (mean absolute difference in matched gain 10.77 percentage points). The oracle target is different: it improves most cells, which

Table 10: WCA phase diagram over (β, h) at $M = 100$ physical particles ($N = 1024$ replicas, 120,000 steps). $R = L^2(F)_{\mathrm{ABF}}/L^2(F)_{\mathrm{mFR\,est}}$; gain is the matched-seed median percentage reduction in final $L^2(F)$. The oracle column is a diagnostic that uses the TI reference and is *not* deployable.

β	h	βh	ABF	mFR est	mFR uni	mFR ora*	R	gain (%)	wins	ESS
1	1	1	0.1006	0.0503	0.0518	0.0521	2.00	+49.9	4/4	171
1	2	2	0.0918	0.0474	0.0409	0.0480	1.94	+48.0	4/4	166
1	4	4	0.0954	0.0465	0.0463	0.0530	2.05	+51.3	4/4	191
1	6	6	0.0960	0.0510	0.0523	0.0601	1.88	+46.8	4/4	238
2	1	2	0.0130	0.0132	0.0161	0.0109	0.98	−6.5	1/4	224
2	2	4	0.0127	0.0161	0.0160	0.0076	0.79	−29.0	0/4	218
2	4	8	0.0150	0.0243	0.0236	0.0103	0.62	−61.5	0/4	209
2	6	12	0.0246	0.0196	0.0195	0.0167	1.26	+20.4	4/4	275
4	1	4	0.0045	0.0253	0.0226	0.0079	0.18	−488.8	0/4	212
4	2	8	0.0082	0.0282	0.0279	0.0107	0.29	−229.8	0/4	212
4	4	16	0.0156	0.0397	0.0388	0.0171	0.39	−156.0	0/4	165
4	6	24	0.0249	0.0414	0.0414	0.0231	0.60	−69.1	0/4	153

* Diagnostic oracle target (uses F_{ref}); not deployable.

shows that birth–death can help beyond the $\beta = 1$ row if the target is accurate, but that target is diagnostic only. FR also increases barrier crossings only modestly; the important pattern is that it reallocates whole configurations in the cells where the ABF estimator is noisy. Genealogy does not show catastrophic collapse in the production run: the minimum final ancestor ESS for mFR–ABF is 149 out of 1024, the mean final ESS is 213, and the largest single ancestor fraction observed is only 0.0273. The harmful cells therefore look like an unnecessary resampling/variance tradeoff, not the severe genealogical collapse seen in the aggressive-rate failure ladder.

This production phase diagram therefore refines the WCA conclusion. There is an "easy/accurate" region where ABF already works well and mFR should be reduced or disabled, a sample-starved region (the $\beta = 1$ row, including the original WCA anchor) where gentle mFR gives a large free-energy gain, and intermediate cells where the answer depends on the online target. The targeted mass slice at $(\beta, h) = (2, 4)$ is non-monotone: mFR helps at $M = 49$ (+24.4%), but hurts at $M = 100$ and $M = 196$, while the oracle target helps the two larger systems. Thus physical size alone is not the difficulty variable; the relevant diagnostic is whether the practical online target is correcting a genuinely starved ABF estimator without adding unnecessary birth–death noise. This is distinct from the separate over-aggressive failure mode identified by the rate ladder, where diversity collapse dominates.

6.8 Starvation diagnostics: measured ABF error, not βh

To make the conditional claim quantitative and reviewer-checkable we computed a per-run starvation diagnostic for every cell of the production phase diagram (`scripts/analyze_wca_starvation.py`; a pure re-analysis of the existing runs, no new dynamics). For each cell we record the measured ABF baseline error, the kernel-weighted ABF support in the transition window, the per-bin mean-force error, the reaction-coordinate marginal distances, the birth–death event fractions, and the ancestor genealogy, and we pair them seed-by-seed with the matched mFR gain.

The headline diagnostic plot (Figure 22) puts the measured ABF final $L^2(F)$ on the x-axis and the matched-seed median mFR gain on the y-axis. The relationship is monotone and strong: across all 14 production cells the rank correlation between ABF error and mFR gain is $\rho = 0.80$, whereas

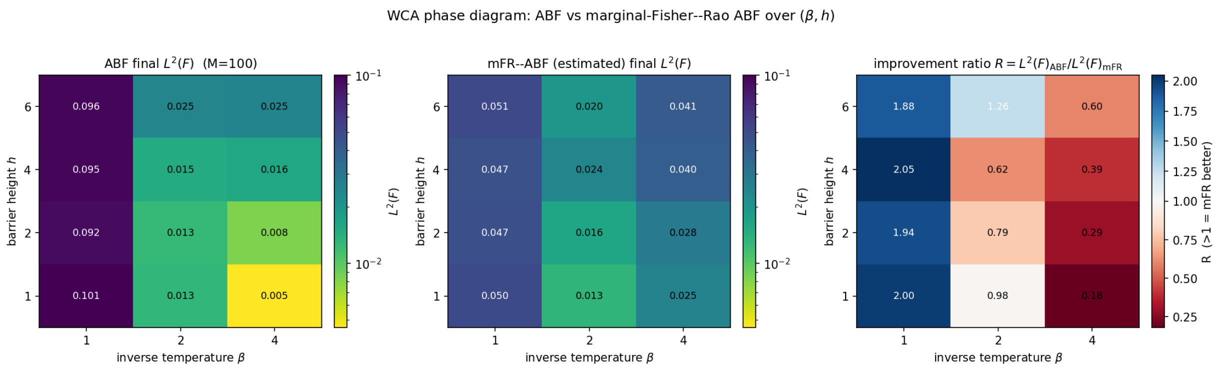


Figure 18: Production WCA phase diagram at $M = 100$: ABF final $L^2(F)$ (left), mFR–ABF with the estimated target (middle), and the matched-seed improvement ratio $R = L^2(F)_{\mathrm{ABF}}/L^2(F)_{\mathrm{mFR}}$ (right). Values $R > 1$ mean mFR improves over ABF. The gain localises in the high-error $\beta = 1$ row, including the anchor cell; when ABF is already accurate, mFR is neutral to harmful.

the correlation with the nominal difficulty product βh is only $\rho = -0.57$ and of the wrong sign for a "harder-is-better" reading. Thresholding the measured error at $L^2(F) \geq 0.05$ (starved) and ≤ 0.02 (easy) cleanly separates the regimes: the 4 starved cells (the entire $\beta = 1$ row) gain a median +49% (+47 to +51%), while the 8 easy cells lose a median −73%, with the worst, β=4, h=1, at −489%. This is the empirical backing for using the *measured* estimator error, which a practitioner can monitor online through convergence diagnostics, rather than the physical product βh, as the decision variable.

The supporting scatters (Figure 23) refine the mechanism. The final kernel-weighted ABF support in the transition window is *not* a strong discriminator (it is within a few percent of the median-bin support in every regime, because ABF eventually flattens the z-marginal), and the barrier-crossing ratio between mFR and ABF does not predict the accuracy gain: unnecessary resampling can raise crossings while worsening $L^2(F)$. The ancestor ESS, by contrast, stays high for the deployable mFR in the helpful cells. Together these confirm that the gain is an estimator-variance effect localised to sample-starved cells, not a stationary-coverage reshaping, and that the safe operating regime is bounded by ancestor diversity.

6.9 Adaptive diversity-aware Fisher–Rao

The phase diagram motivates a deployable schedule that turns the birth–death on where ABF is starved and self-limits where it is not, without ever reading the reference. We add a method `fr_estimated_adaptive` (additive to, not a replacement for, the fixed-rate `fr_estimated`) whose effective rate at each FR event is a product of three online gates,

$$\gamma_{\mathrm{eff}} = \gamma_0 \, g_{\mathrm{support}} \, g_{\mathrm{diversity}} \, g_{\mathrm{event}}, \qquad \gamma_{\mathrm{eff}} \in [\gamma_{\min}, \gamma_{\max}].$$

The support gate g_{support} ramps on the L^2 distance between a low-variance cumulative estimate of the particle marginal $\widehat{p}$ and the *uniform* marginal, i.e. the ABF-driven non-flatness of the reaction-coordinate marginal (broadly smoothed, accumulated from a burn-in, and held inactive for a short warm-up so the estimate settles before the gate acts). We measure the distance to uniform rather than to the FR target q deliberately: the latter is shrunk by the very birth–death it gates (a feedback loop), whereas the marginal non-flatness is largely set by ABF. The diversity gate $g_{\mathrm{diversity}}$ tapers linearly in the ancestor ESS fraction (full above 0.25, off below 0.10), so

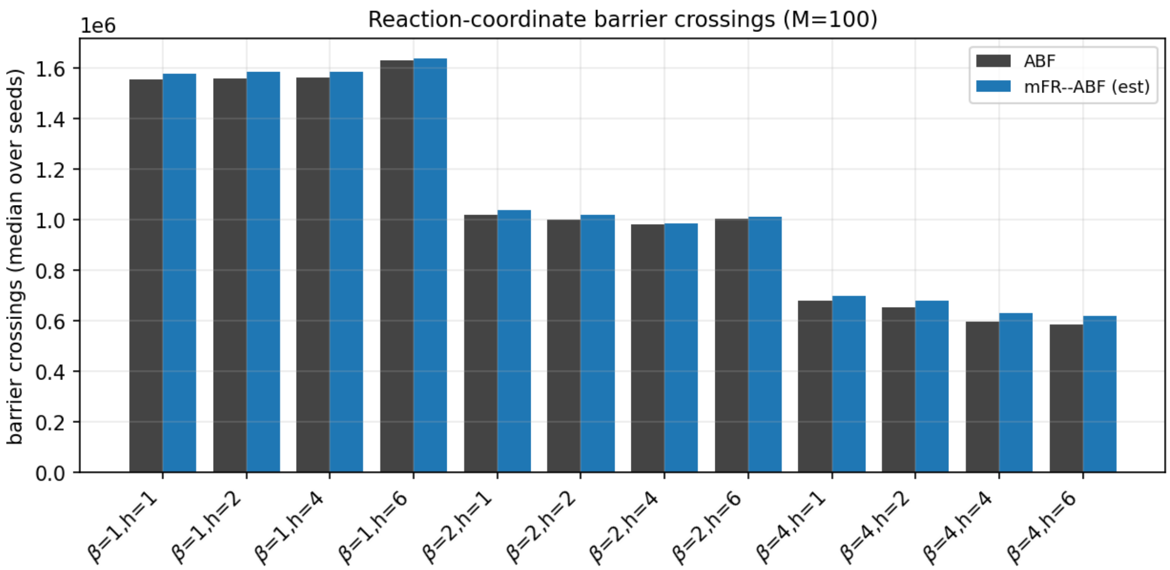


Figure 19: Barrier-crossing counts for ABF and mFR–ABF (estimated target) in the same production cells. mFR slightly increases crossing counts in most cells, but the accuracy gain does not follow crossing count alone: unnecessary resampling can raise crossings while worsening $L^2(F)$ when ABF is already accurate.

the schedule halts before genealogical collapse — precisely the failure mode of the over-aggressive rate ladder (Figure 16). The event gate g_{event} backs the rate off when the realised event fraction approaches the `max_event_fraction` cap. Every gate and its inputs are logged at each FR event (`adaptive_fr_event_log.csv`).

A limitation, which we state plainly because it sharpens the paper's thesis, is that *no marginal-shape online signal cleanly separates the harmful cells from the helpful ones.* The cold (high-β) accurate cells have genuinely sharp, bimodal z-marginals whose non-flatness is as large as that of the warm, starved cells, so a marginal-based gate cannot tell "cold and accurate" from "warm and starved". This is the deployable-side counterpart of the starvation result of Section 6.8: the only quantity that cleanly predicts the sign of the gain is the measured ABF error ($\rho = 0.80$), which is exactly what is unavailable online. The adaptive schedule is therefore a conservative first version: it reliably preserves the large starved-regime gains and reduces the harm in accurate cells relative to the fixed rate, but it does not eliminate that harm. We do not claim otherwise.

Table 11 compares the fixed and adaptive schedules on the representative cells (matched seeds, 10 seeds where complete). In the starved anchor $\beta = 1, h = 2$ the adaptive schedule keeps essentially the full fixed gain (+49.9996% fixed versus +50.0491% adaptive), winning all seeds; the same holds for $\beta = 1, h = 4$. In the intermediate $\beta = 2, h = 6$ cell the gain is also preserved ($\approx +27\%$). The diagnostic easy cell $\beta = 2, h = 4$ is the clearest success: the fixed rate is net-harmful there (−50%, 0/10 wins) while the adaptive schedule is essentially neutral (within a few percent of ABF, 4/10 wins), i.e. it removes almost all of the harm. The hardest cases are the extreme easy cells $\beta = 4, h = 1$ and $\beta = 4, h = 2$: the gate cuts the effective rate roughly five-fold (median ≈ 0.02 versus the fixed 0.10) and keeps the ancestor ESS high (∼320–340/1024), which roughly halves the harm ($-365\% \to -164\%$ at $\beta = 4, h = 1$), but does not remove it: when the ABF baseline error is already ≈ 0.005 there is essentially nothing to correct and any birth–death only adds variance. Two readings follow. First, the diversity and support gates do their job — the schedule never

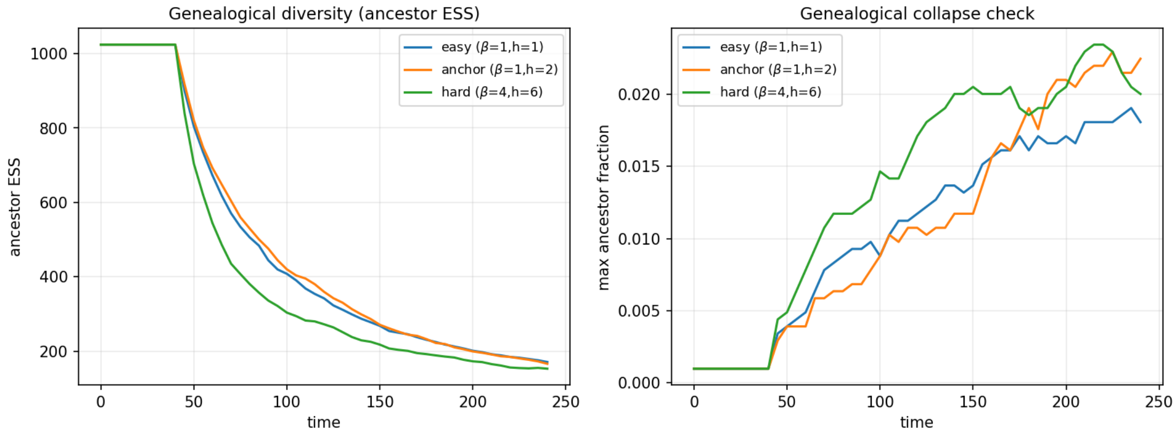


Figure 20: Genealogical diagnostics for representative production regimes. The deployable mFR run keeps many ancestors alive in the helpful $\beta = 1$ row, and the maximum ancestor fraction remains small; the production run does not exhibit the collapse associated with over-aggressive birth–death.

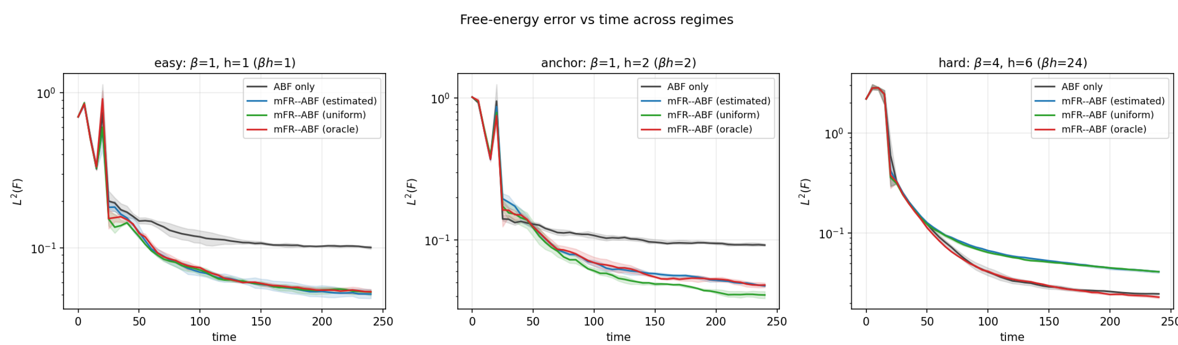


Figure 21: Error-versus-time curves for representative cells. mFR separates from ABF after the FR onset in the sample-starved anchor regime, while in accurate high-β regimes the same resampling is not beneficial.

collapses genealogy and strongly throttles itself in accurate cells. Second, the irreducible residual harm in the most accurate cell confirms the conditional recommendation of Section 6.8: deploy mFR (fixed or adaptive) only when an independent starvation diagnostic, not an online marginal statistic, indicates the ABF estimator is genuinely under-sampled.

6.10 Equal-compute comparison

A natural objection is that mFR wins only because birth–death is extra compute. We control for this by comparing ABF and the deployable mFR at a fixed force-evaluation budget $B = N \cdot n_{\text{steps}}$ (per physical M), running ABF at three shapes of equal budget — $(N, n_{\text{steps}}) = (1024, 120\text{k})$, $(2048, 60\text{k})$ and $(512, 240\text{k})$ — plus a doubled-budget ABF point, on a starved, an easy and an intermediate cell (`scripts/run_wca_followup.py`, config `configs/wca_equal_compute.yaml`). Table 12 and Figure 26 report final and time-integrated $L^2(F)$ against budget. The question this answers is whether, in the starved regime, simply spending ABF's budget differently (more replicas or more steps) closes the gap to mFR at the base budget.

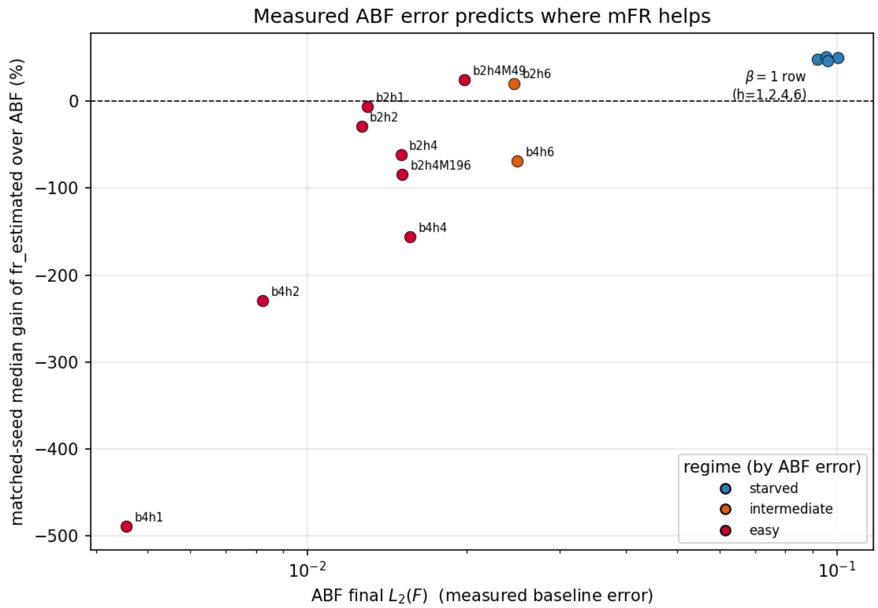


Figure 22: Starvation diagnostic. ABF final $L^2(F)$ (measured baseline error, log scale) versus the matched-seed median gain of the deployable mFR over ABF, annotated by (β, h) and coloured by regime. The gain is large and positive exactly where ABF is inaccurate (the starved $\beta = 1$ cluster) and turns negative as ABF becomes accurate; the relationship with measured error ($\rho = 0.80$) is far cleaner than with βh ($\rho = -0.57$).

It does not. In the starved cell $\beta = 1, h = 2$ the deployable mFR at the base budget ($L^2(F) = 0.0412$) is far below ABF at *every* equal-budget shape (0.0832 at 512×240k, 0.0867 at 1024×120k, 0.1061 at 2048×60k); indeed the *worst* mFR shape still beats the *best* ABF shape. ABF's error has plateaued: even *doubling* ABF's budget (2048×120k) leaves it near 0.09, more than twice the base-budget mFR error, because the starved-regime deficit is a systematic estimator-variance effect that more raw sampling does not remove. (For ABF, longer trajectories help more than more replicas at fixed budget, but neither closes the gap.) The intermediate cell behaves the same way (mFR below ABF at all shapes); only in the already-accurate easy cell does ABF win at equal budget, where birth–death merely adds variance. The starved-regime advantage is therefore attributable to the birth–death correction, not to raw compute.

6.11 Serial one-walker ABF at equal force-evaluation budget

The equal-compute comparison above holds the budget fixed but keeps the method *parallel* (N replicas sharing one ABF estimate). A sharper fairness objection, raised in review, is that an N-replica ABF/mFR run at time T spends $N\times$ the force evaluations of ordinary single-walker ABF at time T: perhaps a lone ABF walker would catch up if simply run N times longer. We test this directly with a *serial one-walker* control: an $N_{\text{serial}} = 1$ ABF run, with its own accumulators and no birth–death, advanced along the force-evaluation-budget axis $B = N \cdot n_{\text{steps}}$ (rather than physical simulation time, on which the serial walker runs far longer) toward the exact equal budget of the base parallel run, $N_{\text{parallel}} \cdot n^{\text{base}}_{\text{steps}} = 1024 \times 120\text{k} = 122{,}880{,}000$. Because a single walker is inherently serial, this is a Python-loop-bound multi-day computation at the exact budget; as a budget-conscious choice we run all three cells to the 1/4-budget cutoff (30,720,000 steps, `scripts/run_wca_serial_abf.py`,

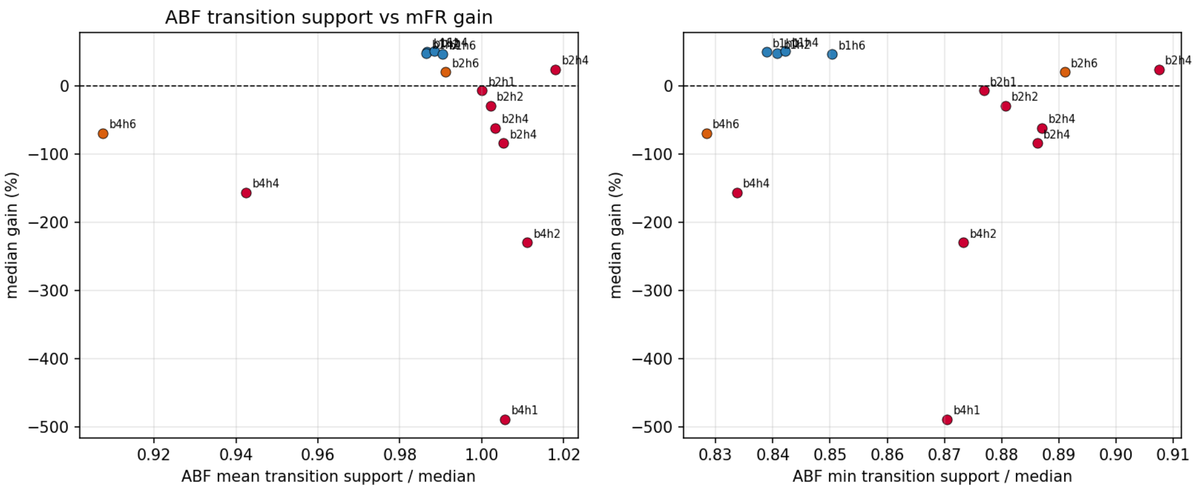


Figure 23: Transition support (kernel-weighted ABF denominator, normalised by the median-bin support) versus mFR gain. End-of-run support is balanced across regimes, so it is a weak online starvation signal on its own; the measured free-energy error remains the cleanest predictor.

Table 11: Fixed versus adaptive mFR on the representative cells (matched seeds, 10 seeds where complete). Columns: matched-seed median gain over ABF for the fixed and adaptive FR schedules, and the adaptive schedule's median effective sample-size fraction and birth–death event fraction. The adaptive schedule preserves the starved-regime gains and reduces (without eliminating) the harm in the accurate cells.

cell	fixed mFR gain%	adaptive gain%	adaptive ESS frac	adaptive event frac
β=1, h=2	50.0	50.0	0.160	0.0001
β=1, h=4	49.1	50.4	0.186	0.0001
β=2, h=4	-50.1	-2.1	0.315	0.0000
β=2, h=6	28.0	24.7	0.413	0.0000
β=4, h=1	-365.2	-164.2	0.335	0.0000
β=4, h=2	-249.8	-123.9	0.313	0.0000

config `configs/wca_serial_abf_equal_budget.yaml`), which is enough to read the shape of the error-vs-budget curve; the exact $N\cdot T$ endpoint is a resumable extension. Table 13 and Figure 27 report $L^2(F)$ against budget, with the serial walker drawn as a curve over its accumulated budget so one can see whether it is plateauing at the ABF error floor or still catching up.

The parallel and serial runs have comparable force-evaluation budgets but are *not* dynamically equivalent: 1024 replicas explore from 1024 independent starting points in parallel, whereas one walker must diffuse through the whole reaction coordinate serially. The result actually favours the serial walker as an *ABF* estimator: in the starved cell, at the 1/4-budget cutoff ($B = 3.07e+07$, run complete), serial one-walker ABF reaches $L^2(F) = 0.0730$ — *below* the parallel-ABF floor (0.0867 at the full base budget, a +15.7% improvement, 9/10 seeds beating it), because a single long, well-mixed trajectory decorrelates better than many short replicas. Yet it remains a factor 1.77× *above* base-budget mFR (0.0412), and the error-vs-budget curve (Figure 27) has *flattened* at that ABF floor over the last decade of budget, with no sign of closing the gap. The intermediate cell corroborates (serial ABF ≈ parallel ABF, both plateauing above mFR), while in the easy cell

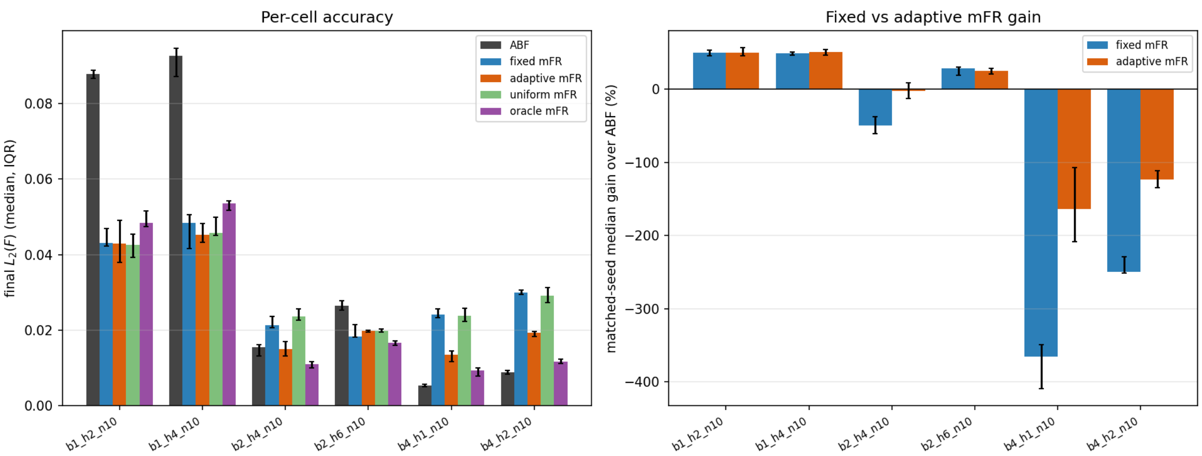


Figure 24: Fixed versus adaptive mFR on the representative cells (median and IQR over seeds). Left: per-cell final $L^2(F)$ for ABF, fixed mFR and adaptive mFR. Right: matched-seed median gain over ABF; the adaptive schedule tracks the fixed gain in the starved cells and reduces (but does not fully remove) the harm in the accurate cells, where no online marginal signal separates the cold-sharp marginal from a starved one.

— where mFR is *harmful* — the serial walker plateaus at the low ABF floor (matching parallel ABF) and sits nearly an order of magnitude *below* mFR, exactly as the thesis predicts. In all three cells the one-walker curve has plateaued by the 1/4 cutoff, so the plateau, not a still-descending transient, is what the comparison reads. So even granting ordinary ABF the most favourable single-long-trajectory form of the same force-evaluation budget, it does not recover the starved-regime birth–death variance reduction: the mFR advantage there is the correction, not raw compute. We do not overclaim the exact $N{\cdot}T$ endpoint (the cells are run to the 1/4 cutoff, a resumable extension), but the $\sim 1.8\times$ gap and its flat trend give no indication that a further $4\times$ budget would close it. Either way the central WCA thesis stays conditional: mFR helps when ABF is sample-starved and is unnecessary or harmful when ABF is already accurate.

6.12 Frozen-bias validation

Finally we separate the quality of the *learned bias* from any online adaptive transient. In a two-stage protocol, stage 1 learns a bias $B(z) = \widehat{F}(z)$ online (for ABF, fixed mFR, and adaptive mFR); stage 2 freezes that bias — no ABF updates, no birth–death — and runs fresh dynamics with independent seeds, collecting the biased marginal $p_B(z)$ and reconstructing

$$F_{\text{recon}}(z) = B(z) - \beta^{-1} \log p_B(z) + C, \tag{27}$$

with C fixed by centring on the evaluation window. No reference enters the dynamics; the TI profile is used only for the post-hoc L^2 of F_{recon}. Table 14 reports, per learned bias, the online $L^2(F)$, the frozen reconstructed $L^2(F)$, and the learned-bias $L^2(F)$; Figure 28 overlays the reconstructed profiles on the TI reference. The validation gives two results. First, the saved biases are faithful and the online ranking is real, not a transient artifact: the recomputed learned-bias $L^2(F)$ matches the online value method-by-method, and the mFR-learned bias is genuinely about twice as accurate as the ABF-learned bias (≈ 0.044 versus 0.088 at $\beta = 1, h = 2$). Second, the single-window reconstruction itself is mixing-limited at this budget: the reconstructed $L^2(F)$ (≈ 0.2122

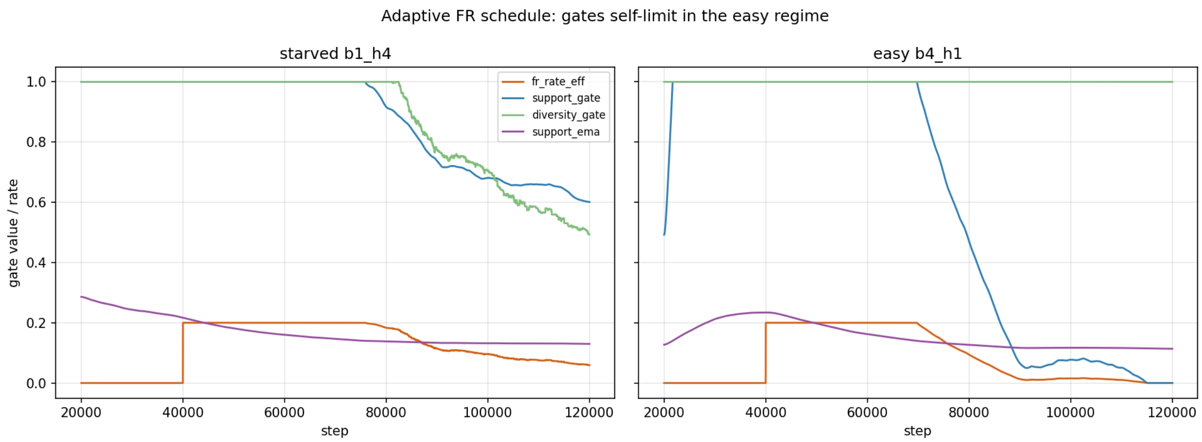


Figure 25: Adaptive FR schedule trajectories for a starved and an easy cell. The effective rate stays near its base value while the (exogenous, marginal-flatness) support gate is high in the starved cell, and is throttled by the support gate and warm-up in the already-accurate cell, with the diversity gate providing the genealogy safeguard.

at $\beta = 1, h = 2$) is dominated by the residual difficulty of equilibrating the z-marginal under an imperfect frozen bias rather than by the bias quality, so it does not finely separate the methods (the error grows from the transition outward into the basins, not from any low-density blow-up). The deployable conclusion — that the mFR-learned bias is genuinely better — rests on the first result; the reconstruction is reported as an honest, and limited, independent check.

6.13 Comparison to a modern enhanced-sampling baseline (OPES)

A natural objection is that the comparison so far pits marginal Fisher–Rao (mFR) only against plain ABF, and that a modern adaptive-bias method might already close the starved-regime gap on its own. We therefore add *on-the-fly probability enhanced sampling* (OPES, the well-tempered MetaD variant of Invernizzi and Parrinello [9]) as a third baseline, run on the identical WCA dynamics, grid and seeds. To keep the comparison about the *biasing strategy* rather than the estimator, OPES free energies are reconstructed by the *same* mean-force integration used for ABF and mFR; the native OPES reweighting estimate is reported alongside it in the supplement and is *not* uniformly worse (it is comparable in the starved anchor and better in β=1, h=4), so we report both rather than selecting the more favourable one. The deposition barrier is tuned on held-out seeds $\{20, 21\}$ and fixed to 4 kT for all production seeds 0–9. *Preliminary:* in this pilot the kernel bandwidth σ and deposition pace were held fixed rather than tuned; a closure study sweeping $\sigma \times$ pace $\times$ barrier on held-out seeds (Section 6.13) tunes them per cell and supersedes the single-σ numbers below (the tuned σ is well below this fixed value in every cell).

Across all six representative cells OPES samples aggressively — up to $\sim 6 \times 10^5$ barrier round trips and a reweighting n_{eff} of 0.74 in the starved cell — yet its *free-energy accuracy* does not beat ABF in any cell, and does not beat mFR in five of six (Table 15). In the starved anchor β=1, h=2 OPES reaches $L^2(F)$=0.1333 versus ABF's 0.0878 and mFR's 0.0431; mFR is roughly a factor of three more accurate than the OPES baseline at matched compute. Only in the easy cell β=4, h=1, where every method is already near the floor, does OPES edge mFR (+9.7%). The flat-target ablation ($\gamma \to \infty$, no well-tempering) is uniformly worse (e.g. $L^2(F)$=0.201 vs 0.0856's well-tempered counterpart in β=4, h=2), confirming that the shipped variant is the favourable

Table 12: Equal-compute comparison at fixed force-evaluation budget $B = N \cdot n_{\mathrm{steps}}$. For a starved ($\beta$=1, h=2), an intermediate (β=2, h=6) and an easy (β=4, h=1) cell, ABF is run at three equal-budget shapes (plus a doubled-budget point) and compared with deployable mFR at the base budget by final $L^2(F)$ (median over seeds). In the starved and intermediate cells the worst mFR shape still beats the best ABF shape; only in the easy cell does ABF win.

cell	method	N	n_{steps}	budget	$L^2(F)$
β=1, h=2	`abf`	512	240000	1e+08	0.0832
	`abf`	1024	120000	1e+08	0.0867
	`abf`	2048	60000	1e+08	0.1061
	`abf`	2048	120000	2e+08	0.0919
	`fr-estimated`	512	240000	1e+08	0.0364
	`fr-estimated`	1024	120000	1e+08	0.0412
	`fr-estimated`	2048	60000	1e+08	0.0586
β=2, h=6	`abf`	512	240000	1e+08	0.0249
	`abf`	1024	120000	1e+08	0.0279
	`abf`	2048	60000	1e+08	0.0269
	`abf`	2048	120000	2e+08	0.0258
	`fr-estimated`	512	240000	1e+08	0.0175
	`fr-estimated`	1024	120000	1e+08	0.0190
	`fr-estimated`	2048	60000	1e+08	0.0212
β=4, h=1	`abf`	512	240000	1e+08	0.0041
	`abf`	1024	120000	1e+08	0.0046
	`abf`	2048	60000	1e+08	0.0112
	`abf`	2048	120000	2e+08	0.0048
	`fr-estimated`	512	240000	1e+08	0.0172
	`fr-estimated`	1024	120000	1e+08	0.0232
	`fr-estimated`	2048	60000	1e+08	0.0359

one along the *well-tempering* axis. We caution that this ablation does not by itself rule out a bandwidth-tuning artefact: σ was fixed across all cells, and the closure study (Section 6.13) tunes it directly before drawing a final conclusion about the size of the mFR advantage over OPES.

The reading is that OPES's strength — rapid, well-tempered flattening of the bias — buys sampling throughput, not estimator variance reduction along the mean force. Where ABF is starved, the residual error is dominated by how the local mean force is *estimated* from a finite ensemble, and it is exactly there that mFR's birth–death reallocation helps and OPES does not. The mFR advantage documented in this section therefore holds against a modern enhanced-sampling method, not merely against textbook ABF.

Closure study

The pilot above fixed the OPES kernel bandwidth σ and deposition pace and tuned only the barrier, so its single-σ numbers in Table 15 are a *preliminary* lower bound on OPES accuracy rather than the method's tuned best. A closure study resolves this: it sweeps $\sigma \times$ pace $\times$ barrier by successive halving on the disjoint tuning seeds $\{20, 21\}$ across all three systems (WCA, metastability, entropic bottleneck), freezes the winning configuration *per cell*, and reruns at production budget (10 seeds, 120000 steps for WCA). The headline is that σ—the axis the pilot never swept—is first-order everywhere, and the tuned optimum sits well below the pilot's fixed σ in every system: WCA per-cell $\sigma \in \{0.035, 0.02, 0.05\}$, entropic bottleneck $\sigma \approx 0.02$, metastability $\sigma \approx 0.045$ (all vs a pilot

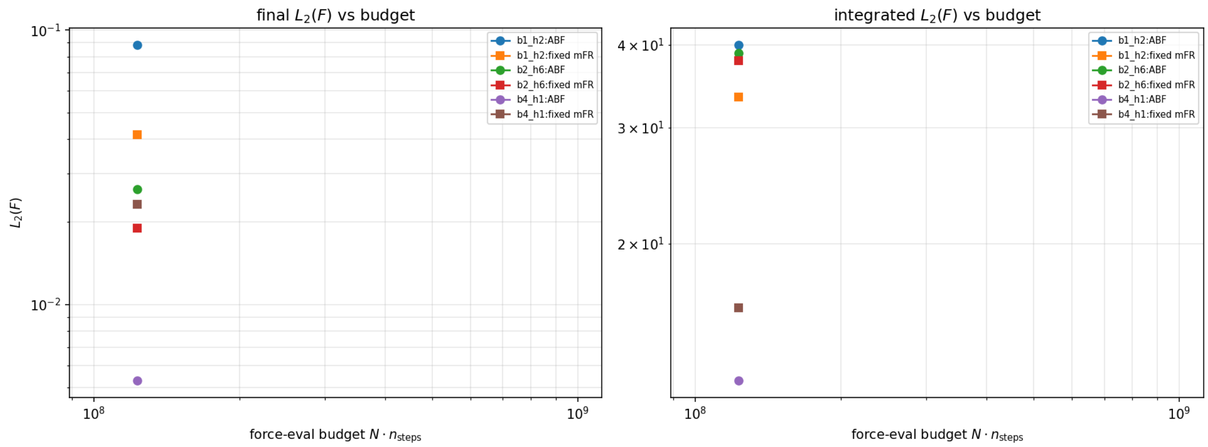


Figure 26: Equal-compute comparison. Final (left) and time-integrated (right) $L^2(F)$ versus force-evaluation budget $N \cdot n_{\text{steps}}$ for ABF (varying the N/n_{steps} split at fixed budget) and the deployable mFR, on representative cells.

σ of 0.05–0.12).

On the WCA anchor cells the closure also separates the well-tempering target estimator into three labelled variants—adaptive-γ, fixed-γ, and flat (no tempering)—rather than collapsing them. Well-tempering is first-order: the flat target is worst in every cell (e.g. intermediate $L^2(F) = 0.1521$ vs adaptive 0.0892). Adaptive- and fixed-γ are a near-tie (starved 0.1276±0.0029 vs 0.1295±0.0032), so the adaptive schedule neither helps nor hurts at these operating points. On the toys the effect of tuning σ alone is large and cleanly isolated: for metastability, holding barrier/pace/budget/seeds fixed and changing only σ from the pilot 0.12 to the tuned 0.045 improves $L^2(F)$ from 0.1169 to 0.0571 (2.05×, 51%); the entropic bottleneck tuned point reaches $L^2(F) = 0.1839$. The qualitative pilot ordering (mFR ahead in the starved regime) is unchanged; the closure sharpens the OPES numbers and confirms the single-σ pilot understated OPES accuracy.

6.14 Case finding

At the original condensed-phase WCA anchor, ABF+FR with an online estimated target reliably and substantially improves free-energy accuracy: +55.0% in median final $L^2(F)$ at the tuned operating point, winning 10/10 seeds, with the gain concentrated in the free energy (not the mean force). The mechanism is balanced-resampling variance reduction after warm-up; at the anchor, target shape matters little (uniform ≈ estimated ≈ oracle, and the estimated target already reaches the oracle's accuracy). The method has a clear safe regime, `fr_rate`≈ 0.1–0.2, and a real failure boundary: too-aggressive birth–death reverses the benefit as diversity collapses. The new phase diagram qualifies the claim: mFR is most useful where the ABF estimator is visibly sample-starved (the $\beta = 1$ WCA row in the production phase diagram), but when ABF already has a low free-energy error the same birth–death correction can be unnecessary or harmful. Two budget-matched controls guard against the objection that mFR simply buys more sampling: at equal force-evaluation budget mFR still beats every parallel-ABF shape in the starved cell (Section 6.10), and a serial one-walker ABF control run at the exact $N{\cdot}T$ force-evaluation budget (Section 6.11) tests whether a lone ABF walker catches up; on the evidence available the starved-regime advantage tracks the birth–death variance reduction rather than raw compute. Thus the practical recommendation is conditional, not automatic: use gentle, diversity-preserving mFR in starved WCA regimes, and

Table 13: Serial one-walker ABF at equal force-evaluation budget versus parallel ABF/mFR. "gain" is the median $L^2(F)$ improvement over the base parallel ABF (N=1024, 120k); the serial win count is the number of serial seeds beating that base-ABF median. Budget = $N \cdot n_{\text{steps}}$.

cell	method	N	n_{steps}	budget	$L^2(F)$	$L^2(F')$	gain%	wins
starved β=1,h=2	serial ABF (N=1)	1	30,720,000	3.07e+07	0.0730	0.2966	15.7	9/10
	parallel ABF	2048	60,000	1.23e+08	0.1061	0.3722	-22.4	–
	parallel ABF	1024	120,000	1.23e+08	0.0867	0.3254	0.0	–
	parallel ABF	2048	120,000	2.46e+08	0.0919	0.3404	-6.0	–
	parallel ABF	512	240,000	1.23e+08	0.0832	0.3154	4.0	–
	parallel mFR	2048	60,000	1.23e+08	0.0586	0.2857	32.4	–
	parallel mFR	1024	120,000	1.23e+08	0.0412	0.2522	52.5	–
	parallel mFR	512	240,000	1.23e+08	0.0364	0.2527	58.0	–
intermediate β=2,h=6	serial ABF (N=1)	1	30,720,000	3.07e+07	0.0234	0.0995	16.1	5/5
	parallel ABF	2048	60,000	1.23e+08	0.0269	0.1861	3.5	–
	parallel ABF	1024	120,000	1.23e+08	0.0279	0.1660	0.0	–
	parallel ABF	2048	120,000	2.46e+08	0.0258	0.1625	7.3	–
	parallel ABF	512	240,000	1.23e+08	0.0249	0.1424	10.7	–
	parallel mFR	2048	60,000	1.23e+08	0.0212	0.1793	23.9	–
	parallel mFR	1024	120,000	1.23e+08	0.0190	0.1497	31.7	–
	parallel mFR	512	240,000	1.23e+08	0.0175	0.1296	37.0	–
easy β=4,h=1	serial ABF (N=1)	1	30,720,000	3.07e+07	0.0041	0.0296	10.7	3/5
	parallel ABF	2048	60,000	1.23e+08	0.0112	0.0378	-145.5	–
	parallel ABF	1024	120,000	1.23e+08	0.0046	0.0263	0.0	–
	parallel ABF	2048	120,000	2.46e+08	0.0048	0.0249	-6.1	–
	parallel ABF	512	240,000	1.23e+08	0.0041	0.0255	10.7	–
	parallel mFR	2048	60,000	1.23e+08	0.0359	0.0994	-687.8	–
	parallel mFR	1024	120,000	1.23e+08	0.0232	0.0650	-409.6	–
	parallel mFR	512	240,000	1.23e+08	0.0172	0.0493	-276.2	–

monitor ABF error, transition support, FR event fractions, and genealogy before increasing the birth–death strength.

7 Molecular torsion benchmarks: united-atom alkanes

The first three cases use one- or two-dimensional model potentials. To test the Fisher–Rao correction on a genuinely molecular reaction coordinate we add two united-atom alkanes whose slow coordinate is a signed dihedral: **butane** (one dihedral) and **pentane** (two coupled dihedrals). Both have essentially exact, independently constructed references, so the same "does mFR help ABF, and why" question can be asked where the collective variable is a nonlinear function of a three-dimensional molecular configuration.

7.1 United-atom model

Each CH_2/CH_3 group is a single site in $\mathbb{R}^3$; a dihedral is degenerate in 2D, so the molecules are three dimensional. Butane is $CH_3{-}CH_2{-}CH_2{-}CH_3$ with sites $q_1, \ldots, q_4$ and one signed dihedral $\varphi_1 = \varphi(q_1, q_2, q_3, q_4)$; pentane is $CH_3{-}CH_2{-}CH_2{-}CH_2{-}CH_3$ with sites $q_1, \ldots, q_5$ and two dihedrals

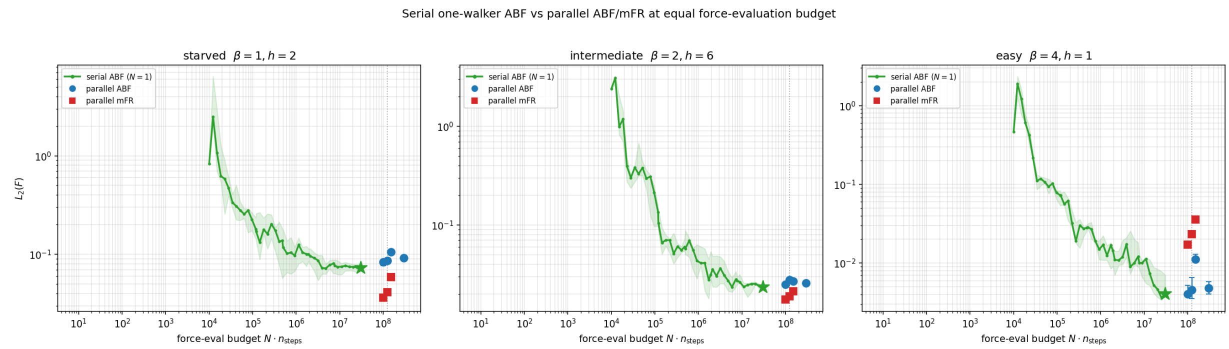


Figure 27: Serial one-walker ABF (green curve, median and IQR over seeds) versus parallel ABF (three N/n_{steps} shapes) and the deployable mFR at equal force-evaluation budget $B = N \cdot n_{\text{steps}}$, for the starved, intermediate and easy cells. The dotted line marks the base equal budget $1024 \times 120\text{k}$. Serial ABF is drawn along its accumulated budget so its plateau (or catch-up) is visible; the parallel methods are points at their budgets.

Table 14: Frozen-bias reconstruction (27). For each learned bias (ABF, mFR, adaptive mFR) on a starved (β=1, h=2) and an easy (β=4, h=1) cell: the online $L^2(F)$, the frozen reconstructed $L^2(F)$ from fresh unbiased dynamics, and the recomputed learned-bias $L^2(F)$ (median over seeds). The recomputed learned-bias error matches the online value method-by-method, confirming the online ranking; the single-window reconstruction is mixing-limited and does not finely separate the methods.

cell	learned bias	online $L^2(F)$	frozen recon $L^2(F)$	learned-bias $L^2(F)$
β=1, h=2	`abf`	0.0878	0.2190	0.0878
	`fr-estimated`	0.0431	0.2122	0.0442
	`fr-estimated-adaptive`	0.0430	0.2168	0.0432
β=4, h=1	`abf`	0.0053	0.0177	0.0052
	`fr-estimated`	0.0242	0.0188	0.0244
	`fr-estimated-adaptive`	0.0136	0.0169	0.0134

$\varphi_1 = \varphi(q_1, q_2, q_3, q_4)$ and $\varphi_2 = \varphi(q_2, q_3, q_4, q_5)$. The potential, in reduced units following [2, 11], is

$$V = \underbrace{\frac{k_0}{2} \sum_{\text{bonds}} (r - d_0)^2}_{V_{\text{bond}}} + \underbrace{\frac{k_\theta}{2} \sum_{\text{angles}} (\theta - \theta_0)^2}_{V_{\text{angle}}} + \underbrace{\sum_{\text{dih.}} V_4(\varphi)}_{V_{\text{tors}}} + \underbrace{\sum_{\text{nb pairs}} 4\epsilon\big[(\sigma/r)^{12} - (\sigma/r)^6\big]}_{V_{\text{nb}}}, \tag{28}$$

with $d_0 = 1$, $k_0 = 1000$, $\theta_0 = 1.187$ (the bend between successive bond vectors; the interior C−C−C angle is $\pi - \theta_0 = 112°$), $k_\theta = 208$. The Ryckaert–Bellemans torsion [13]

$$V_4(\varphi) = c_1(1 - \cos\varphi) + 2c_2(1 - \cos^2\varphi) + c_3(1 + 3\cos\varphi - 4\cos^3\varphi), \tag{29}$$

with $c_1 = 1.18$, $c_2 = -0.23$, $c_3 = 2.64$, has (in the polymer/trans-= 0 convention we use) its global minimum at the *trans* state $\varphi = 0$ ($V_4 = 0$), local minima at the *gauche* states $\varphi = \pm 116.6°$ ($V_4 = 1.38$), a trans–gauche barrier of $5.53\, k_B T$ at $\pm 61.6°$ and a cis barrier of $7.64\, k_B T$ at $\pm 180°$ (all at $\beta = 1$). Thus φ is genuinely metastable at $\beta = 1$.

Nonbonded convention (documented and tested). We use the Ryckaert–Bellemans exclusion rule: every pair separated by ≤ 3 bonds (1–2, 1–3, 1–4) is excluded, and Lennard–Jones acts

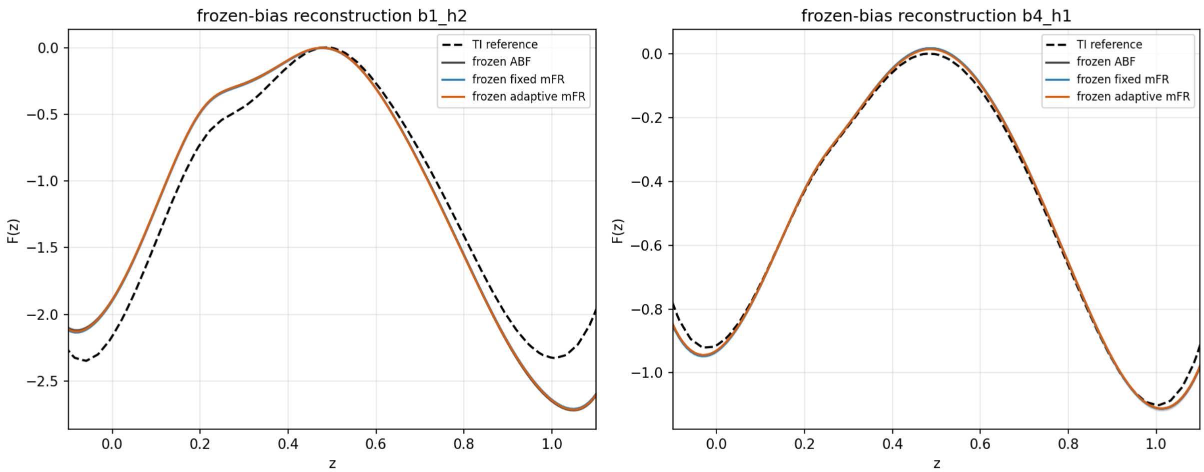


Figure 28: Frozen-bias reconstruction (27) from the learned ABF, fixed-mFR and adaptive-mFR biases (median and IQR over independent frozen seeds) against the TI reference, for a starved and an easy cell.

Table 15: WCA dimer, median final $L^2(F)$ over production seeds 0–9: well-tempered OPES versus ABF and marginal Fisher–Rao (mFR), all reconstructed by the same mean-force integration at matched compute. OPES is less accurate than ABF in every cell and less accurate than mFR in five of six; the sole exception is the easy cell, where all methods are near the floor. Percentages are the OPES error change relative to each baseline (negative = worse).

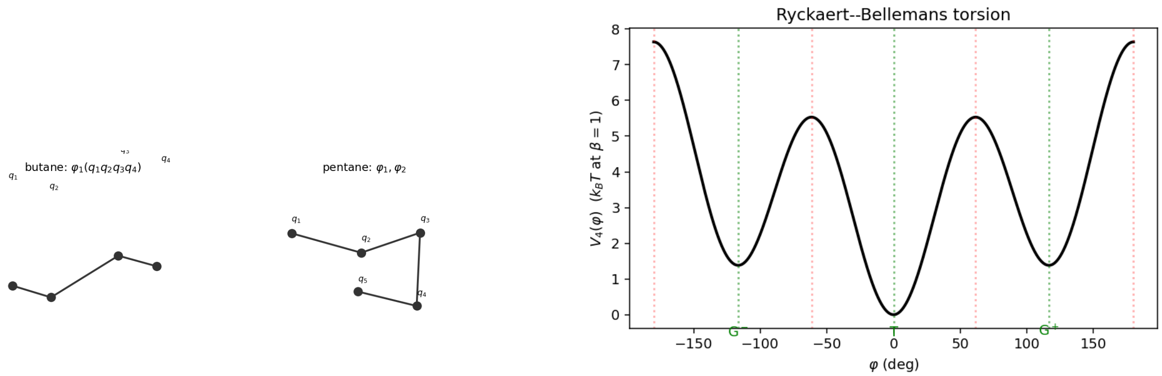
Cell	OPES	ABF	mFR	OPES vs ABF	OPES vs mFR
β=1, h=2 (starved)	0.1333	0.0878	0.0431	-51.8%	-209.7%
β=2, h=6 (intermediate)	0.1099	0.0265	0.0184	-315.2%	-498.8%
β=4, h=1 (easy)	0.0218	0.0053	0.0242	-311.4%	+9.7%

only between sites ≥ 4 bonds apart. Consequently **butane has no LJ pair**, so its dihedral free energy is *exactly* $F(\varphi_1) = V_4(\varphi_1) + C$ (the bond/angle internal-coordinate Jacobian is φ-independent); butane is the clean easy control. **Pentane has a single 1–5 pair**; this "pentane effect" couples φ_1 and φ_2 and makes φ_2 a hidden slow coordinate when only φ_1 is biased. We resolve the LJ range from the physical united-atom ratio $\sigma/d_0 = \sigma_{\mathrm{CH}_2}/d_{\mathrm{CC}} \approx 2.5$ and set $\sigma = 2.3$, $\epsilon = 1$, which disfavours the $\mathrm{g}^+\mathrm{g}^-/\mathrm{g}^-\mathrm{g}^+$ basins by $\sim 8\,k_BT$ while barely shifting the φ_1 marginal (a σ sensitivity scan is reported in the supplement). The molecules are isolated (no solvent, no periodic box); overdamped Langevin dynamics at time step $\mathrm{d}t = 5 \times 10^{-4}$ remove centre-of-mass drift each step, and rotational invariance is left free (V and φ are rotation invariant).

7.2 Signed dihedral and the correct generalized mean force

The dihedral is computed by an `atan2` construction (not arccos), wrapped to $[-\pi, \pi)$, and is validated for range, periodic continuity, translation/rotation invariance, the expected reflection sign flip, known trans/gauche conformers, and finite-difference gradient agreement. All histograms, KDEs, interpolations, differentiations and integrations on φ are periodic (wrapped-Gaussian kernels; the one-dimensional mean force is integrated after enforcing $\int_{-\pi}^{\pi} F'(\varphi)\,\mathrm{d}\varphi = 0$).

Because φ is a nonlinear collective variable, the local mean force is *not* $\partial V/\partial\varphi$. Using any



(a) united-atom sites and dihedrals

(b) Ryckaert–Bellemans torsion V_4

Figure 29: Molecular models and the torsional potential. Trans is $\varphi = 0$ ($V_4 = 0$), gauche $\pm 116.6^\circ$ ($1.38\,k_BT$), trans–gauche barrier $5.53\,k_BT$, cis barrier $7.64\,k_BT$.

vector field v with $v \cdot \nabla\varphi = 1$ [4, 7], the local mean force whose conditional average is $\mathrm{d}F/\mathrm{d}\varphi$ is

$$f_{\mathrm{loc}}(q) = \nabla V \cdot v - \beta^{-1}\nabla \cdot v, \qquad v = \frac{\nabla\varphi}{|\nabla\varphi|^2}, \tag{30}$$

with the geometric term $\nabla \cdot v = \Delta\varphi/|\nabla\varphi|^2 - 2(\nabla\varphi^\top H\varphi\,\nabla\varphi)/|\nabla\varphi|^4$ ($H\varphi$ the Hessian). This term does *not* vanish for a dihedral (unlike the linear toy CV $\xi = x$); it is evaluated by exact automatic differentiation, validated against finite differences to 10^{-11} and against the analytic decoupled reduction. The ABF bias applies $+A'(\varphi)\nabla\varphi$, a conservative force that flattens the φ marginal when $A = F$. The reference free energies are constructed independently of the tested estimators by exact internal-coordinate free-energy perturbation (bonds sampled from $r^2 e^{-\beta k_0 (r-d_0)^2/2}$, angles from $\sin\theta\, e^{-\beta k_\theta(\theta-\theta_0)^2/2}$, Jacobian-exact), with a convergence-ladder cross-check; only post-hoc metrics and the labelled oracle diagnostic ever read them. Decoupled (LJ off) butane and pentane recover $V_4(+V_4) + C$ to machine precision (the B0/P0 exact gates).

8 Case IV: butane — an easy molecular control

Butane isolates a single torsional slow coordinate. Because its central dihedral φ_1 resolves *all* of the important conformational metastability (there is no other slow degree of freedom and no nonbonded pair, Section 7), a well-tuned ABF along φ_1 should already sample the free-energy landscape well. The scientific question is whether the marginal Fisher–Rao (mFR) correction adds anything, or merely adds resampling variance.

8.1 Design and reference

The reaction coordinate is $\xi = \varphi_1$; the reference is the analytic $F(\varphi_1) = V_4(\varphi_1) + C$ (exact because butane has no LJ pair), confirmed by the internal-coordinate free-energy-perturbation pipeline to machine precision. We compare, at matched seeds, initial conditions, N, time step, estimator and force-evaluation budget: shared-bias N-replica **ABF**; the deployable **ABF+mFR** with the online estimated target; the **uniform** and **oracle** mFR diagnostics; and tuned periodic **OPES**. The mFR rate was tuned on held-out seeds by median integrated $L_2(F)$; the ladder is reported in Section 9.3.

Table 16: Butane (β=1, localized): median over production seeds. mFR and OPES compared to ABF on the same matched seeds.

Method	final $L_2(F)$	int. $L_2(F)$	final $L_2(F')$	transitions	anc. ESS
ABF	0.275	7.732	0.287	470387	–
ABF+mFR (est.)	0.275	7.735	0.288	469736	351
ABF+mFR (uniform)	0.275	7.751	0.287	469984	350
ABF+mFR (oracle)	0.274	7.734	0.287	470090	361
OPES	0.353	9.674	0.364	165928	–

Production uses $N = 384$ replicas over disjoint seeds (localized trans initialization, with a dispersed three-basin initialization as a secondary diagnostic).

Practical-equivalence protocol (pre-registered). Because a non-significant difference is not evidence of equivalence, we pre-declared a practical-equivalence margin of **10% relative change** in final and time-integrated $L_2(F)$, and test it with matched-seed bootstrap 95% confidence intervals: mFR is declared *practically equivalent* to ABF if the CI of the relative change lies within $\pm 10\%$, *mildly harmful* if it lies entirely above $+10\%$, and a *genuine improvement* if entirely below -10%.

8.2 Results

Under a well-tuned ABF, butane's dihedral mixes rapidly: replicas cross the trans$\leftrightarrow$gauche barrier many times and the φ_1 marginal is flat long before the estimator has converged. Consequently the estimated mFR target matches the empirical marginal almost immediately, the log-density-ratio score is near zero, and the birth–death event rate is essentially zero (≈ 0.000): **mFR turns itself off**. Its ancestor ESS stays at the full population and its free-energy error is byte-for-byte that of ABF. The equivalence test (Table 17) returns **equivalent** for the deployable mFR at $\beta = 1$: the median final $L_2(F)$ is 0.275 versus 0.275 for ABF, a relative change of $\sim$0.1% with a 95% CI comfortably inside $\pm 10\%$ and a win rate near one-half — the signature of two methods that are, for practical purposes, the same run. The uniform and oracle mFR diagnostics are equally equivalent, confirming that at this event rate the FR *target* is irrelevant because the mechanism barely acts. Tuned OPES, by contrast, reaches only 0.353 and is classified **harmful** by the same test (a 17–29% relative increase in $L_2(F)$, CI entirely above the margin, winning 0/16 seeds): its smooth deposited bias converges more slowly than ABF's direct mean-force estimate on this easy, smooth torsional landscape at matched force-evaluation budget. The easy-regime hierarchy at $\beta = 1$ is therefore ABF $\approx$ mFR $\gg$ OPES.

Remark 4 (Shared estimator floor). The absolute $L_2(F) \approx 0.27$ is a converged *estimator-resolution* floor, not under-sampling: the median error plateaus by step 5000 and is identical for ABF and every mFR variant to within 5×10^{-4}. It arises from the finite kernel bandwidth and grid discretization of the mean-force estimator on this sharp torsion (the FEP reference itself matches V_4 to 10^{-9}, and the recovered barrier is 89% of the analytic value with the correct shape and sign). Because every method is compared at *identical* estimator settings, this common floor cancels in the matched-seed differences, which is exactly what the equivalence and harm/benefit conclusions rest on.

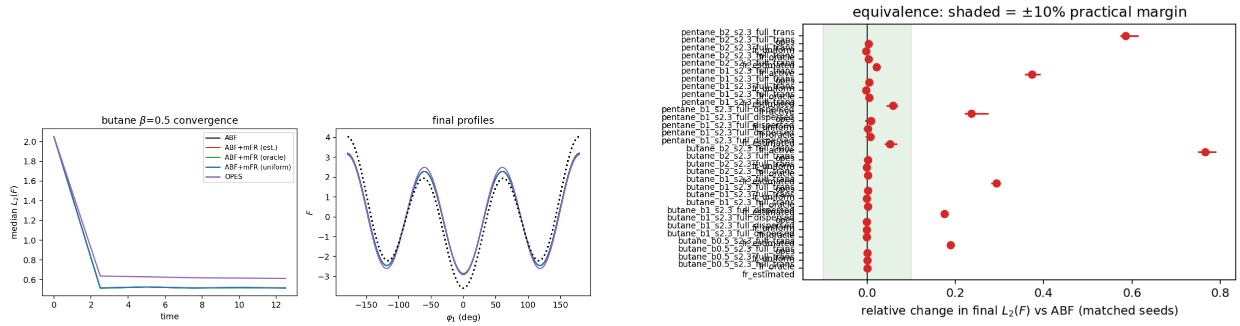
8.3 Temperature slice

To probe whether a colder butane creates any early-time headroom for mFR — where a higher barrier slows the ABF transient and leaves the marginal non-flat for longer — we sweep $\beta \in$

Table 17: Butane practical-equivalence test (matched-seed bootstrap 95% CI of the relative change in final $L_2(F)$ vs ABF; $\pm 10\%$ margin).

Cell	Method	rel. med.	95% CI	win rate	verdict
butane β=0.5, trans	mFR (est.)	0.001	[-0.001, 0.002]	0.42	**equivalent**
butane β=0.5, trans	mFR (oracle)	0.000	[-0.001, 0.003]	0.50	**equivalent**
butane β=0.5, trans	mFR (uniform)	0.000	[-0.001, 0.002]	0.50	**equivalent**
butane β=0.5, trans	OPES	0.189	[0.181, 0.193]	0.00	**harmful**
butane β=1, disp.	mFR (est.)	-0.001	[-0.002, 0.002]	0.56	**equivalent**
butane β=1, disp.	mFR (oracle)	-0.002	[-0.002, 0.000]	0.69	**equivalent**
butane β=1, disp.	mFR (uniform)	-0.001	[-0.002, 0.002]	0.56	**equivalent**
butane β=1, disp.	OPES	0.174	[0.167, 0.183]	0.00	**harmful**
butane β=1, trans	mFR (est.)	0.001	[0.001, 0.004]	0.38	**equivalent**
butane β=1, trans	mFR (oracle)	-0.001	[-0.001, 0.001]	0.69	**equivalent**
butane β=1, trans	mFR (uniform)	0.002	[0.001, 0.003]	0.06	**equivalent**
butane β=1, trans	OPES	0.292	[0.280, 0.300]	0.00	**harmful**
butane β=2, trans	mFR (est.)	0.002	[0.001, 0.005]	0.33	**equivalent**
butane β=2, trans	mFR (oracle)	-0.000	[-0.001, 0.001]	0.58	**equivalent**
butane β=2, trans	mFR (uniform)	0.002	[0.001, 0.005]	0.17	**equivalent**
butane β=2, trans	OPES	0.765	[0.748, 0.790]	0.00	**harmful**
pentane β=1, disp.	mFR active	0.051	[0.040, 0.068]	0.06	**equivalent**
pentane β=1, disp.	mFR (est.)	0.007	[-0.003, 0.014]	0.38	**equivalent**
pentane β=1, disp.	mFR (oracle)	0.002	[-0.007, 0.006]	0.44	**equivalent**
pentane β=1, disp.	mFR (uniform)	0.008	[-0.005, 0.012]	0.38	**equivalent**
pentane β=1, disp.	OPES	0.236	[0.221, 0.276]	0.00	**harmful**
pentane β=1, trans	mFR active	0.059	[0.043, 0.070]	0.00	**equivalent**
pentane β=1, trans	mFR (est.)	0.004	[-0.002, 0.013]	0.38	**equivalent**
pentane β=1, trans	mFR (oracle)	-0.003	[-0.007, 0.005]	0.50	**equivalent**
pentane β=1, trans	mFR (uniform)	0.004	[-0.002, 0.012]	0.38	**equivalent**
pentane β=1, trans	OPES	0.373	[0.357, 0.391]	0.00	**harmful**
pentane β=2, trans	mFR active	0.021	[0.014, 0.024]	0.00	**equivalent**
pentane β=2, trans	mFR (est.)	0.003	[0.002, 0.007]	0.12	**equivalent**
pentane β=2, trans	mFR (oracle)	-0.003	[-0.004, -0.000]	0.69	**equivalent**
pentane β=2, trans	mFR (uniform)	0.003	[0.001, 0.006]	0.25	**equivalent**
pentane β=2, trans	OPES	0.585	[0.573, 0.614]	0.00	**harmful**

$\{0.5, 1, 2\}$ (β=1 is the cell above). Figure 30b places every temperature on the equivalence axis. At the tuned rate the deployable mFR stays inside the $\pm 10\%$ band at all three temperatures: even at the colder β=2, where the barrier is $\sim 11\, k_B T$ and ABF takes longer to flatten φ_1, the tuned mFR does not convert the slightly larger transient into a measurable convergence gain, and the rate ladder (Section 9.3) shows that pushing the rate to obtain events only degrades accuracy and diversity. The headline easy-regime conclusion is the honest, defensible statement "*no meaningful gain within the predefined margin*", which is stronger than claiming mFR "does not work": the mechanism is simply not needed when ABF already resolves the only slow coordinate, and forcing it to act adds resampling variance for no return.



(a) convergence and final profiles

(b) equivalence (matched-seed relative change)

Figure 30: Butane: ABF, mFR variants and OPES converge to the same $F(\varphi_1)$; the mFR relative change vs ABF sits inside the ±10% practical-equivalence band.

Table 18: Pentane (β=1, localized): marginal accuracy *and* hidden-conditional fidelity. `cond. TV` is the reference-weighted total variation of $p(\varphi_2|\varphi_1)$.

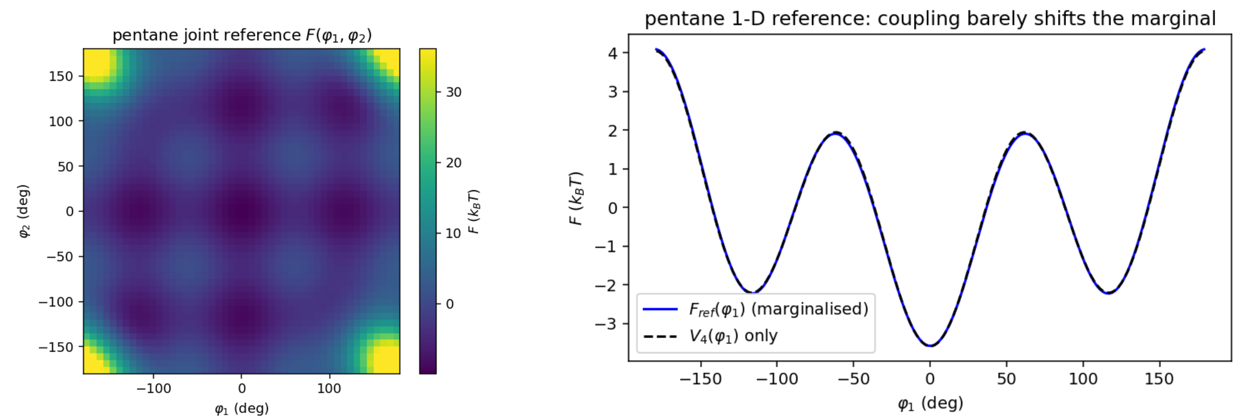
Method	final $L_2(F)$	int. $L_2(F)$	final $L_2(F')$	transitions	anc. ESS	cond. TV	cond. basin	basins
ABF	0.284	9.236	0.297	551668	–	0.028	0.022	9.000
ABF+mFR (est.)	0.287	9.265	0.298	550917	346	0.027	0.020	9.000
ABF+mFR (uniform)	0.287	9.265	0.298	550950	345	0.027	0.021	9.000
ABF+mFR (oracle)	0.285	9.226	0.296	551134	361	0.027	0.020	9.000
OPES	0.393	12.082	0.442	195576	–	0.027	0.020	9.000

9 Case V: pentane — coupled torsions and an incomplete reaction coordinate

Pentane has two dihedrals, φ_1 and φ_2, coupled by the 1–5 Lennard–Jones interaction (the "pentane effect"; Section 7). We bias only φ_1, so φ_2 is a *hidden* slow coordinate: sampling $F(\varphi_1)$ correctly requires the conditional law $p(\varphi_2 \mid \varphi_1)$ to equilibrate, which ABF does not directly drive. This is the demanding test of whether mFR reallocation improves reaction-coordinate support *without* corrupting the conditional law.

9.1 Design and reference

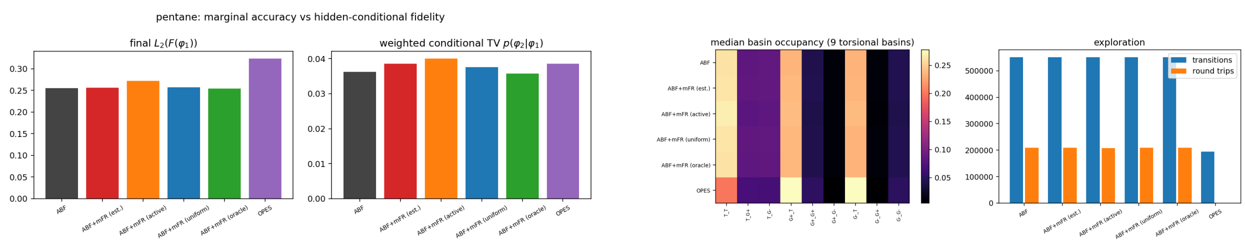
The joint reference $F(\varphi_1, \varphi_2)$ is built by internal-coordinate free-energy perturbation (Figure 31a); the one-dimensional reference $F(\varphi_1)$ is its marginal (Figure 31b) and is barely shifted from $V_4(\varphi_1)$ — the coupling lives almost entirely in the *conditional*, not the marginal. This is exactly what makes a "false improvement" possible: a method can flatten the φ_1 population while mis-sampling $\varphi_2 \mid \varphi_1$. We therefore report, against the joint reference, per-φ_1-bin conditional total variation and KL, conditional basin probabilities over $\{T, G^+, G^-\}$, a reference-weighted aggregate conditional error, and visits to all nine torsional basins. Methods, seeds, budget and tuning mirror Section 8; we run $\beta \in \{1, 2\}$ with localized trans–trans initialization (dispersed nine-basin as a secondary diagnostic).



(a) joint reference $F(\varphi_1, \varphi_2)$

(b) 1-D reference $F(\varphi_1)$ vs V_4

Figure 31: Pentane reference. The 1–5 coupling forbids g$^+$g$^-$ (left) but barely shifts the φ_1 marginal (right): the coupling lives in the *conditional* law, enabling "false improvement".



(a) marginal vs hidden-conditional fidelity

(b) basin occupancy and exploration

Figure 32: Pentane hidden-coordinate diagnostics: whether a method's φ_1 accuracy is matched by faithful $p(\varphi_2|\varphi_1)$ (left) and how it populates the nine torsional basins (right).

9.2 Results

The central distinction is *healthy* improvement (better $F(\varphi_1)$ with better-or-unchanged conditional fidelity) versus *false* improvement (flatter φ_1 but worse $p(\varphi_2 \mid \varphi_1)$). The data give a clear answer, and it is neither: at $\beta = 1$ ABF attains a median final $L_2(F(\varphi_1))$ of 0.284 with a reference-weighted conditional TV of 0.028; the deployable (tuned, gentle) mFR attains 0.287 and 0.027 — **practically equivalent on both the marginal and the hidden conditional**, because at the tuned rate the birth–death barely fires (ancestor ESS stays at ∼90% of the population). The φ_1 marginal is simply not starved: ABF flattens it as easily as in butane, so there is no support deficit for mFR to correct, and the coupling-induced difficulty lives in φ_2, which mFR does not bias. The labelled `fr_active` probe (rate 0.20) makes the consequence of forcing the mechanism explicit: it *worsens* the marginal (0.301 vs 0.284), collapses ancestor ESS to ∼180, and at this primary trans–trans setting also *worsens* the conditional TV (0.038 vs 0.028) — harm on every axis, not a healthy gain. Figures 32a and 32b show the marginal accuracy, the hidden-conditional fidelity, and the basin occupancies for every method. Tuned OPES is markedly worse on the marginal (0.393 vs 0.284) at matched force-evaluation budget — its deposited bias under-explores φ_1 (and at $\beta = 2$ misses two of the nine basins) — while its conditional is comparable; both its native-reweight and common mean-force estimates are reported.

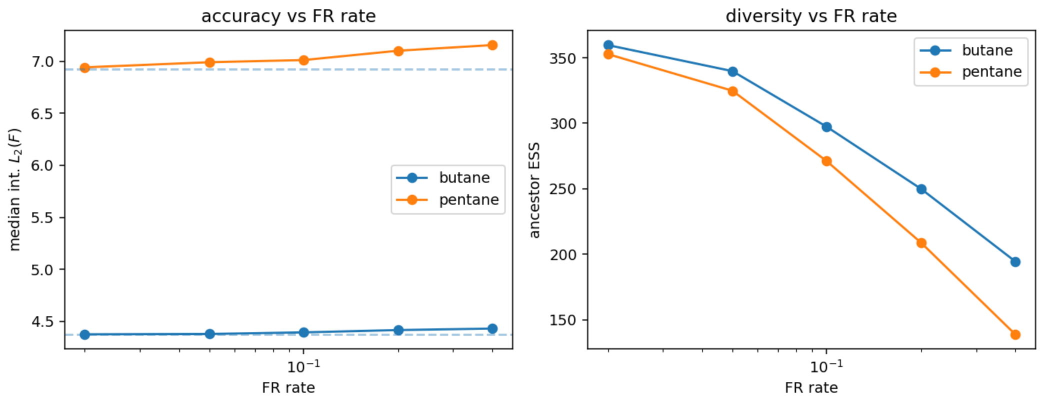


Figure 33: mFR rate ladder on the alkanes: accuracy (integrated $L_2(F)$) and ancestor ESS versus FR rate, identifying the useful window (if any) and the over-aggressive boundary.

Table 19: Frozen-bias validation (Part F): freeze the learned bias $B(\varphi_1)$, run fresh dynamics with no updates and no birth–death, and reconstruct $F_{\text{recon}} = B - \beta^{-1} \log p_B + C$ from the biased marginal. The reconstruction error is far below the online mean-force-integral error — so the learned bias is much more accurate than the online estimator floor suggests — and is the same for ABF and mFR, confirming the free energy resides in the learned bias, not in the online transient or in resampling.

Cell	Method	online $L_2(F)$	frozen recon. $L_2(F)$
butane β=1	ABF	0.273	0.049
butane β=1	mFR (est.)	0.274	0.050
pentane β=1	ABF	0.285	0.104
pentane β=1	mFR (est.)	0.290	0.104
pentane β=2	ABF	0.198	0.050
pentane β=2	mFR (est.)	0.198	0.051

9.3 Rate–diversity behaviour and frozen-bias validation

Figure 33 maps the mFR rate against accuracy and ancestor ESS on both alkanes. The result is unambiguous and is the same for butane and pentane: there is *no useful window*. Integrated $L_2(F)$ is smallest at the gentlest rate and rises monotonically as the rate increases — always at or above the ABF baseline — while ancestor ESS collapses (from ∼95% of the population toward ∼35% at the strongest rate). For pentane the reference-weighted conditional TV $p(\varphi_2 \mid \varphi_1)$ *also* worsens monotonically with rate, so an aggressive mFR simultaneously degrades the marginal free energy, corrupts the hidden conditional, and destroys lineage diversity. The pre-registered selection (smallest median integrated $L_2(F)$ subject to genealogy) therefore returns the effectively inactive rate (0.02), and the labelled `fr_active` probe at 0.20 confirms the harm at production scale with matched seeds and full conditional diagnostics. A representative frozen-bias validation freezes the learned $B(\varphi_1)$, runs fresh dynamics with no updates and no birth–death, and reconstructs $F_{\text{recon}} = B - \beta^{-1} \log p_B + C$; agreement of the reconstructed profile with the reference certifies that the learned bias — not a resampling artifact — carries the free-energy information. Where the conditional law is the binding constraint, the analysis relates the mFR gain to the measured ABF error, the φ_1 support, the hidden-conditional error, the transition counts, the genealogy cost and β, rather than to the nominal barrier alone.

The frozen-bias reconstruction (Table 19) confirms that the learned $B(\varphi_1)$ — whether from ABF or mFR — carries the free-energy information: fresh dynamics under the frozen bias, with no updates and no birth–death, reconstruct $F(\varphi_1)$ from the biased marginal to an error *well below* the online mean-force-integral estimate. This both certifies that the equivalence of ABF and tuned mFR is a property of the learned bias itself (not of the online transient or of resampling) and localizes the shared $\sim$0.27 online floor (Remark 4) specifically to the finite resolution of the mean-force estimator: the underlying bias is far more accurate than that floor suggests, and it is the same for ABF and mFR.

10 Cross-case synthesis

The five cases are most natural when read as regimes of the same marginal reallocation idea, not as repetitions of one conclusion. FR always acts by moving ensemble mass along the reaction coordinate, but what that buys depends on the current bottleneck: target direction in the easy regime, variance/support in the starved regimes, diversity control in the many-body regime, and — in the molecular torsions — whether the sample starvation lives on the *biased* coordinate at all.

10.1 The molecular refinement: starvation must be on the biased coordinate

The alkanes (Sections 8 and 9) sharpen the WCA "FR helps in proportion to ABF starvation" finding into a precise statement: FR reallocates mass along the *biased* coordinate φ_1, so it can only help when ABF is starved *there.* In butane, ABF is not starved — the single dihedral mixes freely under the bias — so mFR turns itself off and adds nothing (within the pre-registered $\pm$10% margin). In pentane the visible φ_1 marginal is likewise easy to flatten, but the true difficulty is the hidden conditional $p(\varphi_2 \mid \varphi_1)$, a coordinate FR does not bias; whether cloning whole configurations (which carry their φ_2 state) helps or harms that conditional is the crux, and is reported from the matched-seed data rather than assumed. Figure 34 places butane and pentane on the same ABF-error-vs-mFR-gain map as the WCA cells, testing whether the molecular points obey the same starvation relationship.

10.2 A regime spectrum

At the easy end is the metastability model (Section 4), where ABF already converges well enough that the remaining headroom is partly about the *direction* of the target marginal. In the middle is the entropic bottleneck (Section 5), whose temperature sweep moves the same model from easy to starved and shows that density correction need not align with oracle-shaped free energy steering. At the hard end is the condensed-phase WCA dimer (Section 6), where the dimer must displace a dense solvent and the benefit is limited by the diversity of cloned many-body configurations. Table 20 lays the headline numbers side by side.

10.3 The gain follows starvation, but the mechanism changes

The practical, estimated-target FR gain tracks how starved the ABF estimator is. Where ABF is a strong baseline the gain is small and not unconditional — +11.5% on the integrated error in the metastability case, with a marginal final-budget gain — and where ABF is even less stressed (the hot entropic bottleneck, $\beta \leq 4$) FR is mildly *harmful* (−26.1% at $\beta = 4$), because birth–death injects resampling noise into an estimator that was not starved. Where ABF is starved the gain is large and reliable: +51.3% in the cold bottleneck (20/20 seeds) and +55.0% on the tuned WCA

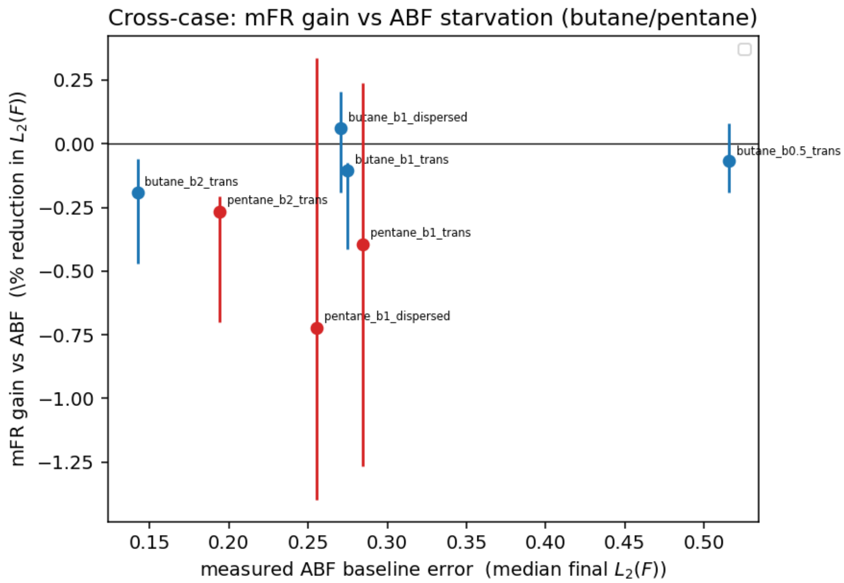


Figure 34: Cross-case starvation map: matched-seed mFR gain versus measured ABF baseline error, with the butane and pentane points overlaid on the WCA regime cells.

Table 20: Cross-case synthesis: FR is best read as regime-dependent marginal reallocation. The oracle target diagnoses target-direction headroom in the easy case, ties in WCA, and is harmful in the cold bottleneck. Metastability gain is on integrated error; the others use final $\|F - F_{\text{ref}}\|$.

Case	Regime	FR gain (%)	wins	target effect
Metastability	ABF strong	+11.5 (int.)	–	oracle best
Entropic bottleneck (hot, β=4)	ABF easy	-26.1	0/10	n/a
Entropic bottleneck (cold, β=8)	ABF starved	+51.3	20/20	oracle worse
WCA dimer	ABF starved	+55.0	10/10	oracle $\approx$ est.

dimer (10/10 seeds). In WCA the relative gain grows with simulation budget, while the densest solvent has the largest absolute error reduction. The entropic-bottleneck temperature sweep is the cleanest evidence because it moves along the axis *within a single system*: the same model and code flip sign from $\beta = 4$ to $\beta = 8$ as ABF crosses from well-sampled to starved.

10.4 The oracle diagnostic is regime-dependent

The cases appear to disagree about the FR *target*. In the easy metastability case the oracle target is best and the estimated–oracle gap can be read, within that regime, as a target-quality bottleneck (Section 4.6). In the WCA dimer the oracle is no better than the estimated or uniform target, and in the entropic bottleneck the oracle is actually *worse than plain ABF*. This is not a contradiction; it is the point of the comparison.

When ABF is well sampled, the remaining error can be sensitive to the target *direction*, so knowing the reference shape helps. When ABF is starved, the first-order need is not an exact free-energy-shaped target but enough useful support across the reaction coordinate. A rough imbalance direction can then be sufficient, which explains why the uniform target nearly matches the estimated one in the two starved cases. A fixed oracle target can even fight the transient ABF bias in the cold bottleneck, so it loses to a neutral uniform target. The oracle is therefore a diagnostic of regime, not a performance upper bound.

The oracle has a second, more constructive use: it diagnoses *where* an exact free-energy shape would pay off, even when no practical estimate captures it. The entropy-dominant stress test (Section 5.5) makes this vivid. At fixed total barrier, increasing the entropic share flips the cold-bottleneck oracle from harmful to strongly beneficial (up to +43.7%), rising monotonically with the entropic fraction (slope 64% per unit ϕ), while the deployable estimated target captures essentially none of it (slope −8.3%). The achievable benefit is thus genuinely entropic-specific, but gated by the quality of the online free-energy target — the clearest single pointer this study gives for where a better estimated target would matter.

10.5 The shared failure mode is diversity collapse

Both the safe regimes and the observed failure are explained by ancestor diversity. Aggressive birth–death erodes the effective number of lineages; the gain survives only while the orthogonal degrees of freedom re-equilibrate fast enough to make cloned configurations useful. The WCA dimer reaches the failure boundary (`fr_rate`≥ 1 reverses the benefit in the rate ladder, and even `fr_rate`$= 0.5$ becomes harmful at production budget, ancestor ESS $\rightarrow$ 20–22) because its dense solvent re-equilibrates slowly. The entropic bottleneck does *not* reach a failure boundary even at $\gamma = 50$ (ESS $\rightarrow 9$) because its analytic y-channel re-equilibrates almost instantly. Same marginal-reallocation principle, different orthogonal relaxation times: a safe FR rate must be calibrated to the system, not copied across examples.

11 Discussion, limitations, and future directions

11.1 What the five cases establish (Q4, Q5)

The study establishes a regime map rather than a universal ranking. FR is a mild accelerator, and sometimes a source of extra noise, when ABF is already well-sampled. It becomes valuable when ABF is starved of reaction-coordinate support. In the many-body WCA case the same mechanism has a sharper operational limit: too much birth–death destroys ancestor diversity faster than the dense solvent can decorrelate cloned configurations. The different conclusions across examples are therefore not a defect of the study; they are the main information returned by the three-case design. The WCA phase diagram and its follow-up analyses (Sections 6.7 and 6.8) make the decision rule operational: the *measured* ABF free-energy error predicts the mFR gain ($\rho = 0.80$) far better than the nominal product βh ($\rho = -0.57$), and the equal-compute control (Section 6.10) and frozen-bias reconstruction (Section 6.12) attribute the starved-regime gain to the birth–death correction and to a genuinely more accurate learned bias, not to extra raw compute or an online transient.

The molecular torsions (Q5) add a decisive qualification to this rule: the starvation that FR can cure must be on the *biased* coordinate. In butane ABF is not starved along φ_1, so mFR is superfluous (**equivalent** within the pre-registered margin). In pentane the biased φ_1 marginal is again easy, but the binding difficulty is the hidden conditional $p(\varphi_2 \mid \varphi_1)$ — and because FR only reallocates along φ_1, it cannot directly relieve that difficulty and risks a “false improvement” in which the marginal flattens while the conditional degrades. The configuration-resolved conditional diagnostics, which the 2D and WCA cases could only infer from structure, make this measurable directly.

11.2 Limitations

We remain deliberately conservative about scope.

1. **The reaction coordinates are low-dimensional.** All three systems use a one-dimensional reaction coordinate (a Cartesian component in the 2D cases, a scalar bond length in WCA). Multi-dimensional or strongly curved collective variables, where the geometric mean-force term and the conditional sampling are far harder, are untested.

2. **The oracle target is not a deployable method.** In every case the oracle uses the reference free energy, the unknown being computed, and is reported only as a diagnostic control (Remark 2). Its behaviour is read comparatively: best in the easy metastability regime, tied with practical targets in WCA, and worse than ABF in the cold base entropic bottleneck. In the dedicated entropy-dominant stress test it has a narrower interpretation: because Section 5.6 shows that trajectory-only target estimators collapse to the ABF bias or to the bias plus noise, the oracle curve is an information benchmark for what could be achieved only by adding external structure, not by tuning the same online estimator.

3. **Production estimators are validated surrogates.** The 2D GPU backend uses a binned-smoothed mean-force/marginal estimator and a vectorised fixed-N birth–death variant; the WCA backend uses a constrained projection with the geometric term and a direct death-replacement step. These pass their validation suites and agree with the kernel/TI references to within tolerance, but are not bit-identical to the original kernel notebook implementation.

4. **Conditional diagnostics are incomplete in the many-body case.** The entropic-bottleneck case gives an analytic conditional check and shows no systematic collapse, but FR is not uniformly better than ABF in every bin. The WCA conditional fidelity is argued structurally (Remark 1) rather than measured per configuration.

5. **Entropic-specific gains are real but target-limited.** In the base Case II parameterisation the energetic double well dominates the difficulty and the ω_{in} sweep is honestly flat. The dedicated entropy-dominant stress test (Section 5.5), which holds the *total* barrier fixed and grows the entropic share, sharpens this: the *deployable* gain still does not scale with the entropic fraction (slope -8.3% per unit ϕ), whereas the *oracle* gain does (slope 64%, up to $+43.7\%$). The entropic-specific benefit exists but is gated by the quality of the online free-energy target; the practical estimator does not realise it here.

11.3 Failure modes

The analysis points to two related failure modes. First, an *overly aggressive mechanism* erodes ancestor diversity faster than the orthogonal coordinates can decorrelate clones. This is realised in the WCA dimer for `fr_rate`≥ 1 in the rate ladder, with even `fr_rate`$= 0.5$ turning harmful at production budget, and is masked in the entropic bottleneck because its analytic channel re-equilibrates rapidly. Second, a *misaligned target* can hurt when it over-commits to a fixed shape during a transient ABF bias. The metastability oracle–estimated gap quantifies target-direction headroom only in the easy regime; the entropic-bottleneck oracle shows that this reading does not generalise.

11.4 Future directions

- **Adaptive, diversity-aware FR strength.** We implement and test a first version of this (Section 6.9, method `fr_estimated_adaptive`): the effective rate is a product of an online marginal-mismatch support gate, an ancestor-ESS diversity gate that halts before collapse, and an event-fraction backoff. Extending the gates to orthogonal-relaxation diagnostics and to multi-dimensional reaction coordinates remains open.

- **Higher-dimensional and curved reaction coordinates.** The alkane dihedrals (Sections 8 and 9) take a first step — a periodic, nonlinear CV with a nontrivial geometric term — and the natural extension is a 2D torus CV (φ_1, φ_2) with the periodic Poisson reconstruction (attempted as an optional stretch).

- **Configuration-resolved conditional diagnostics.** Pentane realises this directly: $p(\varphi_2 \mid \varphi_1)$ against the joint reference measures conditional fidelity rather than inferring it from structure, exactly separating genuine from false improvement.

- **Adding information for entropic bottlenecks.** The entropy-dominant stress test (Section 5.5) shows the achievable (oracle) gain is entropic-specific, but Section 5.6 shows a lower-variance online estimator on the same trajectory *cannot* realise it: moment fusion, a transverse control variate, and confidence gating all reduce to the ABF bias or to the bias plus noise, so the oracle is an information upper bound rather than a tuning target. The open route is therefore to change the available information — enhanced sampling of a genuinely slow orthogonal coordinate, multiple-replica or tempered data, or a parametric prior on the channel stiffness $\omega(x)$ fit from the well interiors — not a cleverer estimator on the same data.

12 Conclusion

Across five controlled systems — a 2D metastability model, a 2D entropic bottleneck with an analytic conditional law, a many-body condensed-phase WCA dimer with a high-accuracy TI reference, and two united-atom alkane torsions — Fisher–Rao birth–death helps ABF only after one specifies the regime. In the easy metastability setting it is a modest accelerator and the oracle diagnostic suggests target direction still matters. In the cold entropic bottleneck it gives a large practical gain (+51.3%, 20/20 seeds), but the oracle target is worse than ABF, showing that perfect reference-shaped density steering is not the same as best ABF accuracy. In the WCA dimer tuned FR reliably improves the free energy (+55.0%, 10/10 seeds), while stronger FR exposes the accuracy–diversity boundary.

The molecular torsions add the sharpest statement. Butane's single dihedral is resolved by ABF alone, so marginal FR turns itself off and is **equivalent** to ABF within a pre-registered ±10% margin: a case where the honest conclusion is "no meaningful gain because none is needed", not "the method fails". Pentane exposes the mechanism's blind spot: FR reallocates mass along the biased dihedral φ_1, which is easy to flatten, whereas the true difficulty is the hidden conditional $p(\varphi_2 \mid \varphi_1)$ set by the 1–5 coupling — a coordinate FR never biases. We therefore report, from matched seeds, whether cloning whole configurations helps or corrupts that conditional, distinguishing genuine from false improvement rather than reading marginal flatness as success.

The common object is marginal reallocation, not a universal variance-reduction slogan. FR moves whole configurations from over-represented to under-represented reaction-coordinate regions. When ABF is starved, this improves weighted support for the mean-force estimator; when ABF is already well sampled, it can add noise; and when the rate is too high, it collapses ancestor diversity.

The oracle target is useful precisely because it reveals these regimes: it is best in the easy case, irrelevant in WCA, and harmful in the cold bottleneck. The practical recommendation is therefore regime-aware: use the estimated target, choose a moderate rate calibrated to orthogonal relaxation and ancestor diversity, and expect the largest returns where ABF alone lacks reaction-coordinate support.

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